

**Quantum machine learning for small datasets:
how far does the quantum advantage hold up in
offering competitive accuracy, execution time, and
practical usability compared to classical machine
learning models?**

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DECLARATION

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ABSTRACT

In 1971, Richard Feynman published the article "Simulating Physics with Computers," where he took the first steps towards building a quantum computer, exemplifying that nature itself is a quantum computer all around us. The future of humanity lies in mastering these machines, because with them we can find cures for diseases at an unprecedented speed, simulate cells and viruses, build materials never conceived, find more efficient ways to produce energy, find alternatives to protect our planet, and enable space exploration. Our science and humanity can take a giant leap forward with these machines, but we encounter problems such as noise and the need to create new quantum algorithms to better explore the quantum properties of nature. This work aimed to analyse new quantum machine learning algorithms, comparing them with classical models running on classical machines, analyse the effects of noise and the ability to deliver solutions, and explore variational quantum circuits that work in a hybrid way on a quantum platform in conjunction with a classical platform.

"The only possible development is human development." (Josué de Castro)

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I want to thank the Irish taxpayer who allowed me to write this master's Thesis. I hope I can always live up to this great opportunity that the Irish people have given me, and I proudly carry the colours of the Irish flag today. A small country, but gigantic in its courage and blessed by great heroes who envisioned its construction in an act of pure altruism, shedding their own blood to guarantee the freedom of future generations!

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LIST OF ACRONYMS

AI	Artificial Intelligence
CN-QCN	China Quantum Communication Network
CNOT	Controlled-NOT gate
COBYLA	Constrained Optimization by Linear Approximation
EPR	Einstein–Podolsky–Rosen (paradox)
FFT	Fast Fourier Transform
GaAs	Gallium Arsenide
IBM	International Business Machines
k-NN	k-Nearest Neighbours
LogReg	Logistic Regression
ML	Machine Learning
MLP	Multilayer Perceptron
NISQ	Noisy Intermediate-Scale Quantum
NIST	National Institute of Standards and Technology
NLP	Natural Language Processing
PCA	Principal Component Analysis
QASM	Quantum Assembly Language
QCL	Quantum Circuit Learning
QFC	Quantum Fidelity Classifier
QFE-LR	Quantum Feature Encoding Logistic Regression
QFT	Quantum Fourier Transform
QKD	Quantum Key Distribution
QkNN	Quantum k-Nearest Neighbours
QMeans	Quantum K-Means
QML	Quantum Machine Learning
QNN	Quantum Neural Network
QPU	Quantum Processor Unit
QRAM	Quantum Random Access Memory
QSVM	Quantum Support Vector Machine
RAM	Random Access Memory
RBF	Radial Basis Function (kernel)
SPSA	Simultaneous Perturbation Stochastic Approximation
SSD	Solid-State Drive
SVM	Support Vector Machine
VQA	Variational Quantum Algorithm
VQC	Variational Quantum Classifier

CHAPTER 1: INTRODUCTION

In 2019, Google announced quantum supremacy with the Sycamore quantum processor, capable of performing calculations in minutes that would take 10,000 years for a classical supercomputer of the time to execute [9]. For many years we have heard about quantum computers and their achievements, such as quantum supremacy achieved by IBM's quantum computer using the Sycamore processor [9]. But what practical applications do these computers have in today's world? Could we use them in daily life, or even for more complex practical problems?

Initially, IBM quantum computers will be used [37], and the models will run using IBM's Qiskit library [16] so that we can analyse under which conditions quantum models can offer practical advantages over consolidated classical models on a small-scale dataset in the current era of noisy intermediate-scale quantum devices called NISQ [17], and whether there is any economic advantage that quantum computers can deliver to motivate the diffusion of their use and operation.

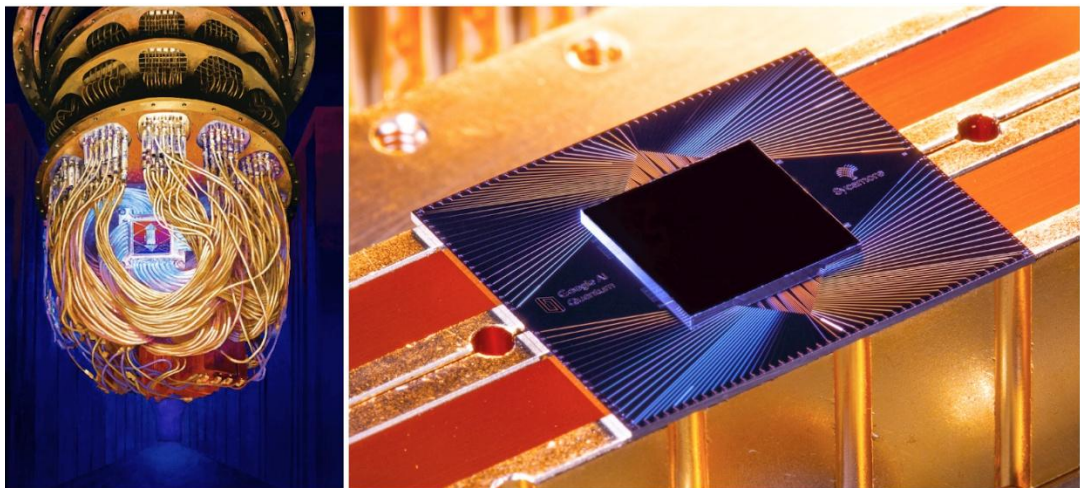


Figure 1: **Left:** Artist's rendition of the Sycamore processor mounted in the cryostat; Forest Stearns, Google AI Quantum Artist in Residence) **Right:** Photograph of the Sycamore processor; Erik Lucero, Research Scientist and Lead Production Quantum Hardware)

After all these years, where are we? We no longer see announcements in newspapers about the advances of these machines and their applications. The roadmap of many large companies in the sector has shown positive results, but there are not many announcements or companies applying this technology in everyday commercial or scientific use that are visible to the public. What happened? Initially, the development of quantum algorithms did not advance as much as hardware development. Quantum algorithms are difficult to implement, and few achieve the performance of Peter Shor's

algorithm [7]. Another important obstacle is the difficulty in guaranteeing reliable results due to the number of errors in quantum processing. Quantum processors are generally susceptible to errors generated by internal noise in the chips, and even when cooled to extremes, task execution generates a small amount of residual heat that interferes with the quantum entanglement of quantum gates, resulting in inaccuracies [17]. This work seeks to analyse current quantum algorithms that can be used in machine learning and natural language processing, comparatively verifying the performance of quantum execution against the computation of the same algorithms on classical computers, as well as analysing the accuracy, cost, and possibility of applying quantum computing in companies. There are potential applications that can be used with quantum computing, such as the simulation of molecules like cytochrome P450, used by the human body in drug metabolism, which has a multidimensional structure that would fit very well with a quantum model that operates within Hilbert space [18], a very large space defined by infinite dimensions.

Although current quantum computers available for use by IBM may have between 152 and more than 500 qubits, the calculations will be limited to only 2 qubits, one for the dependent variable and the other for the independent variable. One of the problems of NISQ computers is noise; the coherence time of qubits is limited to milliseconds, insufficient time for complex calculations, since coherence and quantum entanglement are fundamental pieces in quantum calculations [3]. They are what allow us to use the most significant aspects of these machines. Coherence and quantum entanglement allow the elevation of the number of simultaneous states of operations, enabling results that are difficult to calculate due to problem complexity to be executed. However, our operation based on the Iris dataset will not reach that complexity but rather focus on understanding how a small dataset behaves on a quantum computer. We will focus on quantum clustering using QMeans (Quantum K-Means) [19], [33] equivalent to the classical K-Means model, and on the use of supervised quantum classifiers based on Qiskit and AerSimulator [16]. Classical models will be used as a control so that we can analyse the accuracy of results based on quantum models, offering a benchmark to analyse the gap that could justify the use of quantum computers and low-dimensionality models. The study will allow us to discuss the limitations imposed by noise in the current NISQ era [17].

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Cray Research was a company founded by Seymour Cray in 1972 and its advances enabled much of our current numerical and data processing technology. Advances such as freon refrigeration, the use of integrated circuits, and the great advance of Gallium Arsenide-based semiconductors required a very large investment from the company, almost leading it to bankruptcy due to high research and

development costs [20]. Gallium Arsenide semiconductors have an important characteristic due to their epitaxial growth: cubic crystalline structures are more constant and allow high electron mobility, enabling chips to achieve higher speeds, high radiation resistance making them less susceptible to interference that avoids processing errors, and greater thermal efficiency allowing operation at higher temperatures than silicon. These characteristics allow Gallium Arsenide (GaAs) to work at high frequencies and higher temperature regimes, which were crucial elements for increasing processing capacity for supercomputers used by the US government in calculating simulations of atomic bombs that required massive calculation and data processing. Although Gallium Arsenide could bring many advantages to CRAY, the investment in research to improve this material involved experiments in space where CRAY sent experiments on the space shuttle, which involved huge sums of investment while the world changed the computational paradigm with the WINTEL era, a colloquialism between Windows and Intel that moved the server market. Low-platform computers with Intel's x86 architecture began to operate in parallel and, together with Microsoft's new server operating systems, popularized distributed processing. Large mainframe systems moved away from proprietary architectures and began to work with several Intel processors in parallel with i486 architectures and, at the same time, these computers became the basis for supercomputer processing architectures. With the advance of this new WINTEL paradigm, the end of the Cold War and finally the death of Seymour Cray, the world lost one of the great stars of computer architecture advancement. Cray went bankrupt in 1995 and was sold to Silicon Graphics and finally to Hewlett Packard, which still maintains CRAY supercomputers that are currently sold to research institutions and universities, being used for weather forecasting processing and various health, physics, and another research.



Figure 2: Cray-1 on display at the [Science Museum](https://www.sciencemuseum.org.uk) in London

Big Data was a significant advance in IT, it was now possible to turn used and simple computers into a supercomputer to process large volumes of data. This new data architecture allowed the area of artificial intelligence to become popular and reach companies, and at the same time the intensive use of web technologies with distributed processing, web services, and other hardware architectures such as those from Nvidia could accelerate artificial intelligence areas. The exponential growth of data storage capacity, the increase in RAM memory capacity, and the advent of SSDs allowed companies to make the use of machine learning algorithms fast and easy to incorporate into the core of their business, increasing their predictive capabilities and, along with this, creating new competitive advantages. Companies such as Amazon, Google, Facebook, and several other large players today use predictive algorithms on a large scale and in real time. Investment banks have the capacity to follow the financial market and determine the best investment in real time. All this advance accompanied by new system development methodologies based on agile methodologies that allow several daily updates, considering hundreds of software updates in a single day. Such updates allow the drift of machine learning models so that they can adapt the models to new market and user trends, ensuring competitive efficiency for these companies. Although these technologies are facing great diffusion within the market, being accessible to small and medium-sized companies that, with the adoption of cloud architectures, allow these companies elastic growth of their infrastructure capacities in a fast, secure, and financially compatible way with the

business.

According to Murray Gell-Mann [21, p. 21], "there are more ways for the molecules of a gas of oxygen and nitrogen to mix than to separate. Insofar as things are left to evolve by chance, one can predict that a closed system, characterized by some initial order, will evolve toward disorder, which offers so many more possibilities."

Commonly, entropy reminds us of the idea of increasing disorder or spatial randomness in the universe. We live in this universe with these characteristics, but we act as agents of negative entropy in our universe and where time can be defined as a metric to which we are tied and where the universe begins with order and advances toward disorder. It is important that we highlight the importance of entropy as a central element in all aspects that permeate this work, because in it we can involve the difficulties of building quantum computers with low susceptibility to noise that generate errors in the processing of these machines and also the important aspect that these machines can help build the future of humanity, but it is necessary to understand that the future must be in humanity's control. Ilya Prigogine, in his speech upon receiving the Nobel Prize in 1977, stated that "science for the benefit of humanity is only possible if a scientific attitude is deeply rooted in culture as a whole. This certainly implies better dissemination of scientific information to the public on the one hand, but on the other, a better understanding of the problems of our time by the scientific community" [22, p. 214]

2.2 *Fundamentals of Quantum Machine Learning*

Feynman was a prominent theoretical physicist who participated in the Manhattan Project and contributed to the creation of equations still undisclosed for the calculation of atomic bombs, and he kick-started the discussion on the construction and future use of quantum computers in a 1981 paper [23]. The most interesting part of his paper, which deals for the first time with the discipline of simulating physics in computers, is that the text is a conversation with the reader; when reading the article you are talking to him in a very informal way, as if you were next to him, which captivates the reader and student. But it was Feynman who first said that nature is a large quantum computer, such as the photosynthesis process that over recent years has had important work for the advancement of biological and physical science. Leaves separate oxygen and carbon molecules in highly efficient processes that cannot be explained by classical physics, initially leaving a conclusion that plants use quantum processes to process these molecules, making them one of the main quantum phenomena in nature. Understanding this type of process can lead us to build more efficient quantum machines, because our current quantum computers suffer greatly from noise effects generating errors due to this type of phenomenon that makes quantum computers less efficient than classical computers [17]. Perhaps in the future we can build quantum machines that can help us produce oxygen on the surface of Mars efficiently and practically for future human colonies.

The Connection Machine was an initiative based on the work of Feynman and several other scientists

such as Wolfram (future creator of Mathematica) who together personally engaged in building the first massively parallel processing computer that used multiple processors in parallel. Distributed communication problems were solved by ideas from Feynman himself with teams of Caltech students [25]. Feynman argued that to advance science further we needed machines that could compute massive volumes of data in parallel so that we could simulate complex effects of nature. Over recent years, universities and research centres have increasingly drawn on the results of these initial efforts of the ideas that Feynman sowed, seeking to advance science and its dissemination using computers. The Connection Machine constituted a multidisciplinary effort where Stephen Wolfram helped to conceive, together with Feynman, the machine's hypercube architecture, because the number of connections would reach 10 to the 12th connections; the use of hypercubes simplified the number of direct connections. The second advance for the use of quantum machines was Peter Shor's algorithm that used the Quantum Fourier Transform (QFT) to break encryption keys.

In 1994, Peter Shor and Don Coppersmith discovered how to perform a Quantum Fourier Transform (QFT). A Fourier Transform, when applied to a given signal, returns the frequencies of that signal. If we separate a regular number of values by applying a quantum superposition that repeats in multiples of these regular quantities to these regular spaces, at a certain point the factorization of these spaces reaches values that are multiples, with the division of the factor generated by the encrypted information having a remainder of 1. The fact that QFT can perform this processing in parallel due to the quantum superposition of many qubits separated by frequencies allows the keys to be discovered quickly because they are repeating in frequencies, something that would be impossible for a classical computer to achieve in a short time. Although Shor's algorithm as it is known can be used on classical machines, it presents significant performance on quantum machines [7]. Although quantum machines have problems related to noise, this has been an obstacle for them to increase the number of qubits and consequently leaving the effective use of Shor's algorithm on standby, but we can say that we are close to all world secrets being discovered.

QML (Quantum Machine Learning) as a formalized discipline was recently consolidated according to Biamonte et al. [24], [28], [29]: "offer an influential synthesis of the field, distinguishing approaches that accelerate the linear algebra underlying learning from those that use trainable parametrized circuits." However, significant expenditure on data preparation is required before execution.

A useful tool that greatly assists in preparing and debugging the application before it runs on a quantum computer is AerSimulator, which is an important tool for adjusting the application.

According to the work of Biamonte et al. [24], QML is distinguished by approaches that accelerate parallel linear algebra compared to trainable parametrized circuits, and part of these implementations are within Qiskit's QML libraries and present theoretical gains, making use of efficient data access (QRAM), but which are not always feasible in practice [30]–[32]. The work will test this.

Hilbert space constitutes a useful tool for the use of quantum computing, being multidimensional

spaces that allow the application of quantum models that can be implemented in quantum computing-based applications.

2.3 Quantum Computing

Before we begin to understand quantum computing, it is necessary to speak about quantum theory, which arose at the beginning of the last century, when Max Planck, one of humanity's great physicists, turned his attention to the black-body problem — which we may regard as a hypothetical object that absorbs the electromagnetic radiation incident upon it; in the case of absorption, the radiation is neither reflected nor does it pass through the body. When this body absorbs electromagnetic radiation, it begins to emit electromagnetic radiation that decays over time and can be measured through temperature. We may give a simple example: in steelworks, molten steel emits electromagnetic radiation (light) and, through the intensity of this radiation, we can measure the temperature of the material, which decreases as it cools until it reaches room temperature. This effect puzzled many scientists for years, and one of them was Max Planck, who ultimately resolved the problem by assuming that the energy of the oscillations in the cavity is quantised, and in 1901 he coined Planck's law of black-body radiation [1], [2]. However, some aspects of how the radiation was emitted had not been resolved; this problem drew the attention of a young physicist named Albert Einstein in 1905, who used the concept of a quantised cavity as part of his work on the photoelectric effect, creating the *quanta* of energy that today are called photons [2]. Over the years, Einstein's work was developed by various physicists, culminating in what we now know as quantum theory — even though Einstein was publicly a strong opponent of quantum theories; by life's irony, he was the father of the quantum theory we know today, through his work on the photoelectric effect.

In 1927, the physicist Werner Heisenberg formulated the uncertainty principle, which states the following:

“The imprecision (or uncertainty) in a particle's position multiplied by the imprecision (or uncertainty) in its velocity has a minimum value” [3].

This means that when we try to determine a particle's velocity, we lose the ability to know its position; or, when we manage to measure its position, we are unable to measure its velocity. This principle has a direct application to the double-slit experiment, where quantum effects become most evident: a beam passing through a double slit with a screen behind it produces a wave-superposition effect when the photons strike the screen, classifying light as a wave-like object. But if we place an observer to check through which of the slits the photons striking the screen pass, the photons lose their wave-like character and strike only two points on the rear screen, characterising light as corpuscular. The act of measurement causes the wave to collapse, transforming the wave of the light beam into a photon and characterising it as corpuscular; this wave-particle duality is described by Heisenberg's uncertainty principle and is the starting point for understanding how quantum computers work, since they essentially exploit nature's capacity to operate in a state of uncertainty [3], [4].

2.3.1 Quantum Superposition

Within the Copenhagen interpretation, quantum superposition is regarded as non-realist, because before a particle's state is measured it holds several superposed realities within a mathematical formalism — the Schrödinger wave equation — making it a non-deterministic state; when the particle's state is measured, the wave equation collapses, reducing the particle's state to a single state [3], [4]. The collapse of the wave equation during measurement brings the particle into our reality, whereas before it is superposed across several simultaneous states; quantum superposition is a non-deterministic feature, unlike what classical physics presents regarding the state of particles. Bringing this to a *qubit*, the state of quantum superposition allows the *qubit* to assume several simultaneous states at once; in the case of bits, we can say that a *qubit* holds state 0 and state 1 at the same time. This type of behaviour is the main feature of quantum computers and what makes them such powerful machines. Although it may seem out of the ordinary, quantum superposition is extensively corroborated by the formalism of quantum mechanics over these more than 100 years of theory, having undergone numerous laboratory tests and everyday use in electronic devices [3].

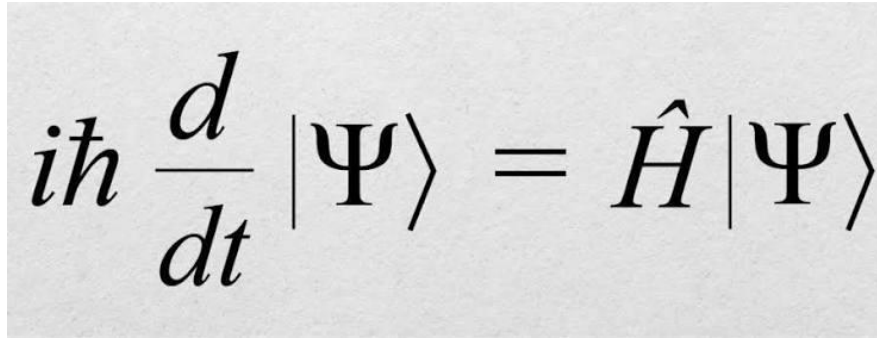

$$i\hbar \frac{d}{dt} |\Psi\rangle = \hat{H} |\Psi\rangle$$

Figure 3: Schrödinger equation describes the temporal evolution and spatial behavior of quantum systems, functioning as the fundamental pillar of non-relativistic quantum mechanics.

2.3.2 Quantum entanglement

Quantum entanglement was called by Albert Einstein “spooky action at a distance” in his famous paper *Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?*, published by Albert Einstein, Boris Podolsky and Nathan Rosen — which became known as the EPR paradox (the initials of Einstein, Podolsky and Rosen) — in the journal *Physical Review* in 1935 [4], [5]. The paper argued that quantum entanglement broke the principle of non-locality, which is the property of light travelling through space independently of the velocity of its source. Entanglement is a physical phenomenon in which two particles, photons or electrons, are connected, where the state of one determines the state of the other regardless of the distance between them. Entangled particles may possess a communication between them that is independent of the speed of light — which is precisely the reason for the EPR paradox: changing the state of one particle changes the state of the other entangled particle faster than the speed of light [4]. This principle of change is a subject of study in quantum cryptography, since, with the particles entangled, they require no direct communication, creating an ultra-secure communication channel; it is also used as a principle of quantum computers [3].

“In 2020, China established an integrated space–ground quantum communication network, comprising 32 backbone nodes and 4 metropolitan networks. Here, we present

China's latest progress in deploying QKD networks, designated the China Quantum Communication Network (CN-QCN). The CN-QCN is an operational, long-haul, trusted-relay QKD network spanning more than 10,000 kilometres, incorporating 145 optical-fibre backbone nodes and 20 metropolitan networks, covering 17 provinces and 80 cities" [6].

Quantum coherence is the property of a particle, or electron, of maintaining multiple states or locations simultaneously without being affected by external disturbances. Only particles or electrons in a state of quantum coherence can maintain quantum superposition and quantum entanglement [3]. When a qubit suffers a disturbance and loses its quantum state of superposition or entanglement, it enters a state of quantum decoherence, meaning the loss of quantum behaviour. Only qubits in a coherent state can be effectively used in quantum processing; disturbances produce unexpected results and introduce errors into the process, generating inaccuracies. The decoherence state is a collapse of the wave function, and it is welcome only now of measuring the results of quantum processing, measurement being a controlled action within quantum computers [3].

2.3.3 Quantum Coherence

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2.3.4 Qubits

Qubits are the fundamental units of quantum information and computation; they may be compared to classical bits, but they operate in quantum superposition, a behaviour impossible in a classical bit. Quantum superposition allows a qubit to hold several simultaneous states at once, being 0 and 1 simultaneously, and these can be combined through quantum entanglement with other qubits. The power of quantum superposition and quantum entanglement gives the quantum computer a data-processing capacity that cannot be surpassed by classical computers [3, p. 8].

2.3.5 Quantum Gates

Quantum gates are like the gates we have in classical computers, but they can operate in quantum superposition and entanglement, thereby allowing us to make combinations of gates. This combination ensures that quantum computers can generate combinations of results that are impossible on a classical computer. Among the main gates, we have the Hadamard gate, responsible for placing a single qubit into superposition, and the CNOT gate (controlled-NOT), which connects qubits in parallel, generating combinations of entangled qubits and producing the quantum entanglement effect [3, p. 17].

2.3.6 Josephson Junction

Brian Josephson was a British theoretical physicist, winner of the 1973 Nobel Prize for his discovery of the effect that bears his name, the Josephson effect, which is based on the quantum tunnelling effect [8]. Quantum tunnelling is the ability of certain particles to cross barriers that were considered impassable by classical physics; this is due to the wave-like effect based on the Schrödinger wave equation, which probabilistically predicts that the particle will appear on the other side of the barrier even without having enough energy to cross it [3]. This effect would be like throwing a ball at a wall, which would normally bounce back, but the ball passes through the wall. The Josephson effect was observed in a junction of two superconducting electrodes at temperatures close to absolute zero, separated by an insulating material; the effect is a macroscopic quantum phenomenon, rather than one on the atomic scale, and can be observed and measured [8]. Josephson junctions are widely used in quantum computers to generate qubits and quantum gates, being a fundamental building block in the construction of current quantum computers [3], [8].

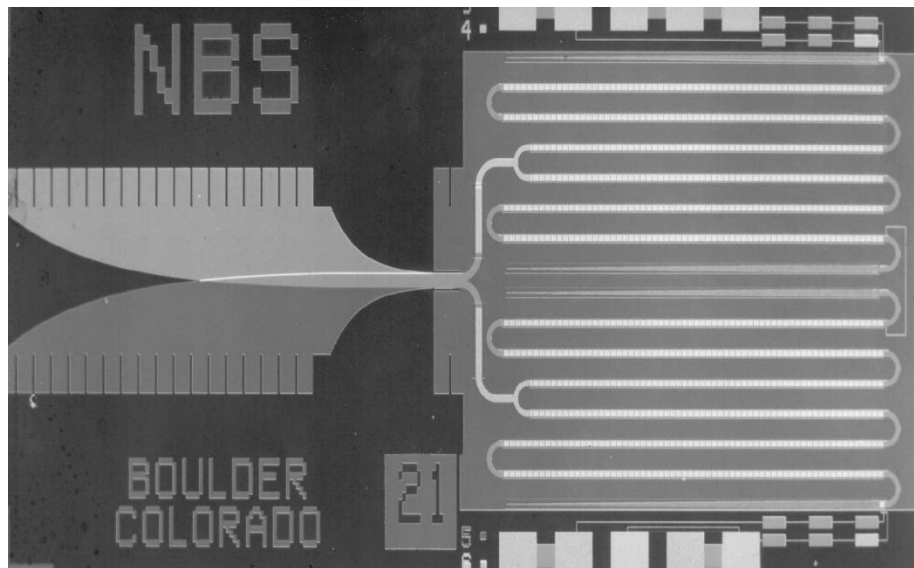


Figure 4: Josephson junction array chip developed by the [National Institute of Standards and Technology](#) as a standard volt

2.3.7 Quantum Algorithms

One of the first quantum algorithms was Shor's algorithm [7], developed by the mathematician Peter Shor and which, ironically, had the same objective as the first computer built by Alan Turing — to break cryptographic keys — constructed at Bletchley Park during the Second World War. In Shor's algorithm, the breaking of cryptography is achieved through the factorisation of numbers with the aid of the Fast Fourier Transform (FFT); to date, it is one of the few quantum algorithms able to exploit the performance of quantum computers significantly [3], [7]. Although the roadmap of quantum computers has been delivering results, the field of quantum algorithms has not advanced at the same pace. We still face the issue of noise, which makes quantum computers less precise than classical computers — something explored in various works. The use of algorithms based on variational algorithms, developed to reduce errors, still relies on classical computers for the processing and correction of the data [3]. New technologies for building quantum chips are advancing, and significant gains in quantum coherence time are increasing year on year — one of the main elements preventing the large-scale adoption of quantum

computers, since the longer the coherence time, the more possibilities there are for implementing complex applications that use these computers' quantum computational power [3].

2.3.8 Types of Quantum Computers

We must clarify that there are currently two classifications of quantum computers: the first by computational architecture — the way in which processing problems are solved — and the second by the type of physical technology used, which is directly related to the manipulation of qubits

2.3.8.1 Classification by hardware type — Qubits

Superconducting: these are computers based on qubits that operate cooled to sub-zero temperatures, producing superconducting effects that eliminate electrical resistance and allow electrical currents to flow without losses. To achieve this, extreme cryogenic refrigeration technologies are currently used, which increases the complexity and cost of machines using this type of technology. Google and IBM have been investing over the years in this type of quantum-computer architecture.

Trapped ions: the ions are held in a vacuum and being electrically charged, are controlled by magnetic and electromagnetic fields and manipulated by laser beams. They offer high precision in operations requiring long coherence times, owing to the stability of this model. However, the operations are slower and the complexity of increasing the number of qubits poses significant engineering challenges.

Neutral atoms: these are like trapped-ion technology regarding control and manipulation, but the difference is that the atoms are neutral. This allows a significant reduction in unexpected interactions and the grouping of thousands of atoms into microscopic arrays organised and optically controlled by lasers known as optical tweezers. This type of technology allows greater scalability and is considered a promising technology for the development of quantum computing.

Photonic: as the name suggests, this technology uses photons, allowing operation at room temperature without the need for extreme refrigeration technologies, photons being natural particles for the transmission of quantum information [36].

2.3.8.2 Classification by computing architecture

Universal Quantum Computer: It is currently one of the models with one of the most popular quantum architectures on the market, being very close and analogous to classical computer architecture. As a universal quantum computer, its architecture aims to solve any type of problem. The idea of the universal quantum computer is based on a quantum Turing machine, initially proposed in an Oxford University journal by David Deutsch, who proposed the use of quantum gates that could function similarly to the binary logic gates of a classical computer. This model has been implemented by IBM, Google, Microsoft and IonQ.

Analogue Quantum Computer: D-Wave was the first company to present a commercial quantum computer and has adopted the analogue quantum-computer architecture, a specialised and restricted model based on the mapping of specific problems, ideal for simulations in chemistry and physics, with a focus on significant performance gains.

Quantum Annealing: This is a quantum-computer architecture focused on optimising complex, specific combinatorial problems of the NP-hard type, and which suffers less impact from the noise normally found in universal quantum computers — an architecture that allows the use of a larger number of qubits in its design.

2.3.9 What Has Been Achieved

To date, one of the greatest problems we face is not having a noise-free universal quantum computer, which profoundly affects the adoption of this technology. Noise is directly related to the coherence time of qubits, a crucial time that needs to be long for certain calculations that require a longer entangled state; with shorter entangled states, more complex results cannot be achieved. Below is a summary of what has been achieved in recent years.

2.3.9.1 Error correction

In 2024, Google, using its Willow chip with 105 qubits, demonstrated for the first time an exponential suppression of errors below the surface-code threshold: the more physical qubits used to form a logical qubit, the lower the error rate. Quantinuum, which uses an ion-trap architecture, managed to produce multiple logical qubits with fidelity higher than that of the physical qubits, able to perform logical operations between them. IBM has an error-mitigation roadmap around 2029 and has, over the years, been advancing in mitigating the errors generated by noise.

2.3.9.2 Quantum supremacy

In 2019, Google announced quantum supremacy to the world using its Sycamore quantum processor [9], an announcement contested at the time by IBM a few days later, which argued that its Summit supercomputer could perform the same calculations in 2.5 days rather than 10,000 years, so that the 200 seconds taken by Google's quantum computer would not represent a definitive supremacy [10]. According to IBM, the order of magnitude between the two computers represented only a temporary advantage until its supercomputer could reach the same performance. An important aspect to highlight in these quantum-performance tests is that they are carried out on quantum circuits that have no practical application and, consequently, are not used to solve major real-world problems.

2.3.9.3 Practical applications

One area that has been gaining attention in quantum computing is the simulation of materials and molecules; because these elements have multidimensional structures, the software complexity increases greatly, requiring many states to be simulated simultaneously. Quantum computers are machines that work within the multidimensional Hilbert space, which guarantees a processing capacity that could hardly be achieved by classical computers. This kind of possibility opens opportunities for simulating new drugs and cellular behaviour for testing new medications. D-Wave has been striving to present solutions based on machine learning optimisation for logistics and transport, in processes where annealing-based computers have a differentiated processing capacity [11].

2.3.9.4 Quantum networks and cryptography

In 2016, China launched its first quantum-communication satellite, called Micius (Mozi), named in honour of the Chinese philosopher Mozi. In 2017, its first quantum communication took place, in which

China demonstrated quantum key distribution. An important aspect to highlight is that quantum communication is a form of cryptography, since there is no way to interfere with the communication without the entangled particles losing quantum coherence. China is today at the forefront of quantum communication and has been planning the launch of additional satellites in its QSS (Quantum Secure Satellite) constellation.

2.3.9.5 Democratisation and software

Currently, all major cloud-service providers offer access to quantum computers, with a minimum amount of time available for using and testing these machines, as well as compiler technologies, mostly accessible through these providers' documentation. Examples of new languages and libraries that have been advancing over the years in quantum programming are Qiskit, Cirq and PennyLane, in addition to the provision of simulators that can run locally on the developer's machine or in a distributed manner on cloud instances, using video accelerator cards that allow significant performance in simulations.

2.3.10 What We Have Not Yet Achieved

In 2025, HSBC announced a 34% improvement in the probability of success in winning clients in the European corporate-bond market, in partnership with IBM [12]; the Brazilian bank Itaú announced that it is exploring the use of quantum computing in its financial operations [13]. Although the announcements are optimistic, the truth is that we have nothing significant in the commercial or scientific use of quantum computers; at present we have no major discovery based on the use of these machines, much less their everyday commercial use with results that would justify their large-scale deployment. The promise of the machines is still very great, but the results are still small and unachieved. It is true that the hardware has been advancing greatly, but there is a large gap on the algorithm side, even though the tests carried out in the present work demonstrated that classical computers still deliver more reliability and performance than quantum computers. Considering Shor's algorithm [7] and Grover's algorithm, they are the ones with the best performance for using quantum-computing resources.

2.3.11 Main Challenges

2.3.11.1 About hardware, the main problems to be solved are as follows.

Better qubits: qubits with lower error rates will be able to produce ever-better results. For this to be achieved, future technologies must be less susceptible to noise; one of the causes of this noise is the proximity of the qubits, which end up leaking electrical signals and heat through the internal substrate of the material in which they are built. The proximity of the qubits interferes with other nearby qubits, generating decoherence effects.

More qubits: with the reduction of noise effects, it is possible that the number of qubits will increase and, consequently, the computational capacity of quantum computers, with significant impacts on the cost of building new, more performant machines.

Cooling and supercooling: quantum computers based on superconductivity require lower temperatures close to zero. Improvements in cooling processes have a significant impact on the cost of quantum computing; computers with optimised cooling and supercooling systems can make the physical

structure of quantum computers more efficient. Better cooling systems can directly increase the coherence time of qubits, which is directly related to our ability to process more complex models that depend on prolonging the qubits' quantum properties.

Error correction: error correction depends on many kinds of solutions. One of the alternatives currently used is the use of several qubits as if they were one, self-correction and the use of error-correcting algorithms; another example is the use of variational quantum circuits, which employ quantum and classical computing resources in a hybrid manner for error correction.

2.3.11.2 In software, the main challenges to be solved are as follows

Quantum algorithms: quantum algorithms are recent and have been receiving much optimisation effort; however, they demand a great deal of practical and theoretical work to evolve.

Hybrid quantum–classical computing: quantum computers depend on classical computers to function; integrating them in an optimised way brings important benefits to various areas of quantum computing, mainly in error correction and in implementing more complex algorithms capable of simplifying adaptation to quantum computers, which function differently in many respects from classical computers.

Error mitigation: more sophisticated software can predict and help reduce errors, which may allow quantum computers to become more precise and efficient, reducing the use of dedicated hardware for error correction.

2.3.11.3 People

Scientists and researchers: although initiatives such as those of IBM and D-Wave have been dedicating important efforts to disseminating quantum technologies, in both software and hardware, there is a need for investment in research at universities and for greater access to the machines. This requires more professionals in the field so that it can develop as an area of technological and scientific advancement capable of improving the adoption of quantum computing as an important field for human development.

Software developers: not all quantum computers are compatible; there are differences, although there have been significant advances in transpilers and adaptable libraries that allow systems to be migrated between different quantum computers. Having a universal language helps greatly in training new developers. Although there are software resources for development on quantum computers, the truth is that the technology and the understanding of it are not trivial and demand a more robust education in quantum-physics concepts; in this regard, we fall short in training developers. To disseminate the technology, more professionals developing on a day-to-day basis are needed, and this requires a lower usage cost, greater availability and greater accessibility to these machines.

Business analysts and end users: *these are the final element in the adoption and use of quantum computers, responsible for making the most of all the resources these machines can bring to the sciences and to commercial use.*

2.3.12 The Iris dataset

The Iris dataset is a multivariate dataset used and popularized by the British statistician and biologist Ronald Fisher in his 1936 article entitled "The use of multiple measurements in taxonomic problems as an example of linear discriminant analysis" [26]. The present work aims to explore the use of quantum

computers using the Iris dataset with both the simplest and most sophisticated resources for machine learning applications, comparatively using classical models, quantum models, and variational quantum circuit-based models that seek to improve the accuracy of these quantum systems [27], [34], [35]. The trials will collect processing time, task execution complexity, and mainly the cost of execution.

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2.5 CONCLUSION

The aim of this work is to make a modest effort in the use of quantum computers. From the very moment Richard Feynman set out the possibilities of these machines [14], he opened a multitude of development possibilities for future generations, based on improving our technical and scientific capabilities. Quantum computers can help us develop new drugs to cure diseases that afflict humanity, simulate human cells in real time so that we can simulate medications to cure cancer, develop technologies that can make weather forecasting far more accurate, develop life-support technologies for space exploration and for future journeys and colonies on Mars, and create new, more efficient and powerful energy technologies. The possibilities are immense for the use of quantum computing, but there are still many steps to be taken; the physical, technological, financial and human barriers hold back these efforts, but it is undeniable that there are incredible prospects for the future of this technology.

When the first computer began operating at Bletchley Park, its sole function was decryption; it is ironic that our first fully quantum algorithm was born with the same objective decades later, through Peter Shor [7]. It is true that there are now new algorithms that even work in a hybrid fashion, between quantum and classical computers, to optimise specific algorithms. Over the years, the roadmap of quantum-computing companies has demonstrated robust progress, but more scientists and researchers focused on improving this technology are needed — a technology that can bring us many benefits; meanwhile, advances in algorithms have not kept pace with the same roadmap as the machines. It would be valuable for quantum physics to be disseminated in schools and universities, so that future students can use these machines more efficiently — quantum physics is not trivial and demands a higher level of abstraction, which would need to be introduced into our schools.

In recent years, the discipline of machine learning and, more recently, natural language processing have been attracting great interest in commercial and scientific fields. Although machine learning has been a discipline adopted in science for many years, in recent years it has received greater impetus thanks to the advancement of new technologies such as Big Data, the reduction in hardware costs and cloud technologies that encourage the adoption of these disciplines. New algorithms and libraries for the use of machine learning and natural language processing are available for use on quantum computers. Although there is, in the present work, no example of natural language processing, machine learning models were analysed with the aid of the Iris dataset, which was the object of testing. What was found was that the use of quantum machines, regarding the models and machines used, did not present

significant gains in accuracy and performance compared with the simulations carried out locally using the Qiskit simulators on a local machine, making it a deterrent to the use of quantum machines. Even so, this study does not represent a nail in the coffin of quantum computing, but rather the starting point for larger studies that may use these machines in a meaningful way, as presented in this work.

Quantum mechanics is one of the most intriguing and impressive disciplines of the natural sciences, being one of the theories most extensively tested and validated in laboratories and universities around the world over its more than 100 years, and one that has been used in many everyday solutions for humanity, including commercial ones. Our inability to understand it lies perhaps in the evolution of our brains, which, by the principle of survival, deprived us of the ability to interpret this fantastic reality around us. As Richard Feynman remarked, nature is a great quantum computer, carrying out quantum processing all around us [14]; the leaves of plants have been carrying out quantum photosynthesis processes for millions of years and at room temperature, something we still do not clearly understand. It is possible that we may significantly advance our understanding of nature using quantum computers.

Kurt Gödel's incompleteness [15] gives us hope that we may advance our understanding of quantum theory soon, thanks to our brains' ability to transcend the limitations that nature, or our own brain, possesses within the interpretation of our reality.

CHAPTER 3: RESEARCH METHODOLOGY

The objective of this chapter is to describe, in a reproducible and comprehensive manner, the methodology used for the empirical investigation of the performance of classical and quantum machine learning algorithms using the Iris dataset. The research characterization investigated questions and hypotheses, and experimental performance based on the computational environment using a classical computer and IBM quantum computing hardware with 158 qubits and the basic development tool of IBM's Qiskit test packages are presented. The specifications and evaluation instruments, the experimental protocol, and the reproducibility and validity control strategies faithfully replicate the computational implementation developed in the `iris_qml` application (available on GitHub), ensuring traceability between the method and the software artifacts. Quantum computing is a promise that needs to be fulfilled, and to achieve this, it is necessary to invest efforts and initiatives so that great results can be realized.

Classification criteria	Framework adopted
Nature	Applied research
Problem approach	Quantitative
Objectives	Exploratory and descriptive
Technical procedures	Experimental, supplemented by bibliographic research.

Classification criteria	Framework adopted
Collection environment	Computational laboratory (local simulation) and quantum cloud (IBM Quantum Platform)
Unit of analysis	Classification model (classical or quantum) evaluated on Iris

Table 3.1 — Methodological characterization of the research.

3.1 RESEARCH METHODOLOGY OVERVIEW

The nature of the work is defined as applied research aimed at generating and improving knowledge of the practical use of quantum computing and currently available technologies based on the latest QML (Quantum Machine Learning) packages, seeking to analyze which machine learning models have a measurable advantage in a classification task using the Iris dataset as a basis. The idea is to use classical machine learning models running on a local machine as controls, and the research uses a quantitatively supported approach and systematic data collection with performance and computational cost metrics that are subsequently treated with descriptive statistics and pairwise comparison. The tests presented the use of variational quantum circuits that use a hybrid approach combining quantum and classical computers, being a completely exploratory and descriptive view focusing mainly on the performance and accuracy of the results. The work also presents data related to times, but they were not the focus of comparisons in this work. The work analyzes the effects on the dependent variables, in this case accuracy, time, and depth of the quantum circuits.

Richard P. Feynman in 1981 in his paper (Simulating Physics with Computers) [1] predicted that quantum computers could perform calculations impossible to be calculated by a classical machine, because due to the quantum capabilities of superposition and entanglement these calculations would be processed very quickly. Four decades later, has this prediction come true?

The machines have become a reality, companies like Google, Amazon, Microsoft, IBM... have advanced research lines in the development of quantum computers and many already make their computers commercially available on the market and are accessible via cloud services. IBM's quantum computers were used in this endeavor, with a 156 Qubit processor running at IBM's Washington site, the company provides 10 minutes of free use for testing its quantum computers. Our tests are a precision comparison between classical computers and quantum computers running machine learning algorithms. Variational quantum algorithms were incorporated into the tests, which perform error correction for the comparative analyses. Several problems were faced head-on: the availability of these computers and the limited time available for test execution. To make the most of the short time available, the same machine learning tests were performed for both classical and quantum algorithms using the AER Simulator quantum simulator to debug all the problems encountered. Since an IBM quantum computer was used, the Qiskit package was employed, which runs on Python and allows both simulation and real-world execution within the quantum platform. The QML (Quantum Machine Learning) package was used in conjunction with COBYLA [3], a classical optimizer package for finding the best parameters for Variational Quantum Algorithms (VQAs) [2].

3.2 Computational Elements and Environment

The quantum architecture used to perform the task was the IBM platform based on the Heron r2 QPU (Quantum Processor Unit) with 133 or 156 qubits; in the specific case of the tests, QPUs with 156 qubits available at the IBM data center in Washington, USA, were used. The following quantum computers were available for this test:

Computer	Quantum Processor Unit (QPU)	Qubits
ibm_kingston	Heron r2	156
ibm_fez	Heron r2	156
ibm_kingston	Heron r2	156

Table 3.2 — IBM Quantum Computers available for use.

IBM Quantum Computers utilize superconductor-based Quantum Processor Units (QPUs) selected in real-time through a configuration in the configuration file (.env) via the variable (IBM_BACKEND_NAME). Qiskit has a less cumbersome server selection feature in the QiskitRuntimeService package, in addition to automatic computer selection and other parameters available in the configuration file (.env) such as backend (for a specific backend choice), credentials, number of shots, transpilation optimization level, variational hyperparameters, and hardware guards. The libraries used in the project are shown below.

Component	Reference Version	Function in the experiment
Qiskit [4]	1.3–1.x	Construction and transpilation of quantum circuits
qiskit-machine-learning [5]	0.8.x	VQC, QSVC, SamplerQNN, fidelity kernels
qiskit-ibm-runtime	0.30+	Submitting jobs to QPU (SamplerV2)
qiskit-aer	0.14+	Local simulation and device noise models
scikit-learn [6]	1.3+	Classic models, metrics, PCA, and scalers.
NumPy [7] / pandas [8]	1.24+ / 2.0+	Numerical and tabular manipulation
matplotlib [9] / seaborn	3.7+ / 0.12+	Generating comparative visualizations
python-dotenv	1.0+	Loading the typed configuration from the .env file.

Table 3.3 — Software stack and its functions.

3.3 Dataset and Pre-Processing

In 1936, the British biologist Ronald Fisher, who was also a well-known statistician, popularized the Iris multivariate dataset in his article, "The use of multiple measurements in taxonomic problems," a classic example of linear discriminant analysis [10]. The Iris dataset was initially a data collection made by Edgar Anderson with the aim of quantifying the morphological formation of iris flowers of three related species, two of which were collected on the Gaspé Peninsula. The Iris dataset is widely used in teaching and machine learning testing and was used in this work. Iris contains 150 samples distributed across three species: Setosa, Versicolor, and Virginica. The samples are organized into morphological attributes: length, sepal width, and petal width. Preprocessing was based on four deterministic steps as described below.

3.4 Standardization

The four attributes of the Iris dataset are standardized using the StandardScaler from SciKit-Learn [6], in this case a Z-score normalization which is a statistical mean of standard deviation better known as a standardized score. The StandardScaler removes the mean and scales to the unit variance $x' = (x - \mu) / \sigma$, this type of standardization is necessary for the PCA and angular coding subsequences which are sensitive to attribute scaling.

3.4.1 Dimensionality reduction and mapping to qubits

The use of PCA [11] was used to reduce the number of dimensions from 4 to 2, in this case reducing the Iris dataset to 2 components, which ended up discarding information and bringing the Versicolor and Virginica classes closer together, however it penalized other models, so much so that later in the confusion matrix we can see that QkNN confused Virginica with Versicolor. The objective of using PCA was to reduce the number of quantum qubits to 2 (NUM_QUBITS, default 2), which decreased the dimensionality of the quantum circuit while decreasing the risk of noise during processing. Thus, enabling execution on NISQ hardware and ensuring a fair comparison between paradigms, the little remaining processing time available for execution on IBM quantum computers made it impossible to incorporate more qubits, as there would be a risk of not being able to finish the execution.

3.4.2 Angle coding

The main components, in this case the Qubits, are rescaled to the interval $[0, \pi]$ using MinMaxScaler, to prepare angle encoding, instead of working with a numerical value. The use of angles avoids periodicity errors where a Qubit vector might be ambiguous with another, as can be seen in the Bloch Sphere [12] below which represents the Qubit.

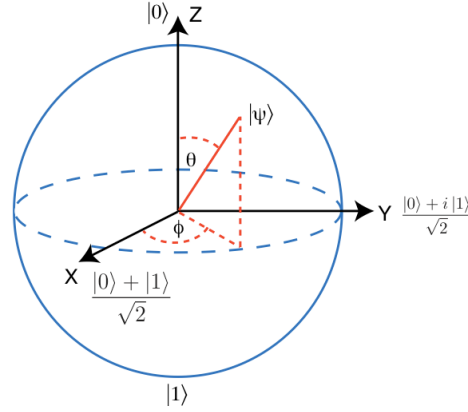


Figure 5. Bloch Sphere representing a single Qubit state.

Bloch Sphere Single Qubit states that are not entangled and lack global phase can be represented as points on the surface of the Bloch Sphere, written as

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\varphi} \sin\left(\frac{\theta}{2}\right) |1\rangle$$

Rotations about the x, y, z axes of the Bloch sphere are represented by the rotation operator gate [12].

3.4.3 Stratified partition

The 80%/20% standard was adopted for sample stratification for training and testing, defined in the TRAIN_SPLIT variable in the ".env" configuration file. Stratification respected the three classes in all partitions, avoiding degenerate partitions, and setting a random seed (RANDOM_SEED, default 42) ensures the exact reproducibility of the split.

3.5 Probabilistic Machines

Quantum computers are by nature probabilistic machines; each execution performed on a set of circuits defined to solve a problem generates a superposition of results—in this case, several results in a given state. However, we cannot extract this information by measuring all these superimposed values at once. Each measurement reads only a single result because the act of measuring the result generates a wave collapse effect, where the values collapse into a single result. The solution to access the remaining results is to execute the programmed quantum circuit again and take the reading. When we execute 1024 shots, we are running the quantum circuits 1024 times and measuring, or rather reading, the result one by one in each round. In this measurement process, we accumulate the results; some are discrepancies due to the nature of the noise generated within these quantum computers. Noise can be generated by heating or proximity of qubits within quantum processors, and by external electromagnetic interference, even when isolated from the environment; therefore, advanced cryogenics processes have a profound impact on these computers.

3.6 IBM Quantum Platform

IBM's cloud platform offers quantum computers with up to 156 qubits, with free plans for beginners who want to learn and use these machines, providing up to 10 minutes of free time per month. This time is sufficient for testing, and it is also possible to use quantum simulators locally before executing definitive tests on physical quantum machines [13]. In addition to the platform, there is a complete learning environment in Qiskit, a framework that allows the use of these computers and is integrated with the Python language, simplifying the experience. Along with this, there is a set of machine learning libraries called QML (Quantum Machine Learning) that run in Python with Qiskit [5]. Together, they will be used to analyse algorithms for the development of the master's dissertation.

For benchmarking, the Iris dataset was chosen, whose wide use in academic literature [10] makes it an adequate reference for this analysis where classical and quantum algorithms will be applied. In addition, new variational quantum algorithms will be used to ensure the accuracy of results [2]. Noise during processing has been one of the major bottlenecks in the use of quantum computing; the use of these algorithms can guarantee better performance in quantum computer benchmarks.

3.7 Quantum Machine Learning

IBM, like many other quantum software and hardware vendors, has been improving its machine learning libraries over the years. IBM, in turn, has a comprehensive set of machine learning libraries called QML (Quantum Machine Learning) [15]. Although the name quantum might sound intimidating, the use of these packages doesn't stray far from the machine learning model used in the most popular packages like PyTorch, SciKit-Learn, Panda... However, one of the main problems that penalizes the use of these algorithms is the lack of QRAM (Quantum Randomic Memory) for storing quantum results, which prevents the use of more complex and lengthy applications, a direct consequence of the problems faced by NISQ (Noisy Intermediate-Scale Quantum) systems [14].

For small datasets like the Iris dataset used in this work, runtime performance left much to be desired. Although the work was not focused on runtime performance, the times were measured and compared to the local computer running SciKit-Learn. In only one case did the processing time of the quantum machine learning algorithms perform very close to that of the local machine, in this case a Dell Precision notebook with an Intel i7 processor, 32 GB of RAM, and 1 Terabyte of SSD. The other results were not promising, as we will present throughout the chapters.

3.8 Variational Quantum Circuits

These were the circuits that performed worst in processing time and did not yield significant results for the Iris dataset tests, as will be presented and analyzed in the test results. Variational quantum circuits are quantum circuits with adjustable parameters, in this case an ansatz (a word that comes from German and means "initial attempt" or "approximation", in this case input parameters in a quantum circuit). Variational quantum circuits work in a hybrid mode, operating in quantum architecture and classical architecture, the training process takes place on a classical computer, being the basic building block of Variational Quantum Algorithms [2]. In the tests carried out, processing in variational circuits demanded

more processing time on the quantum computer due to the local training of the model that was then sent to the quantum computer.

To understand what a variational quantum circuit is, it is necessary to understand what a Hamiltonian is. A Hamiltonian in physics is the sum of all the energies of a system, in this case the sum of the kinetic energy and the potential energy. Based on this information, we can say what can happen to a system based on this information, its future and its past. In quantum computing we can Let's define it as the quantum circuit of the application. Based on this quantum circuit, we have two actors: one validating a response and the other calculating responses. Since we are still in the NISQ (Noisy Intermediate-Scale Quantum) era, quantum computers don't have enough qubits for complex calculations, and a lot of noise is generated due to heating effects and the environment where the quantum computers run. The precision of quantum systems isn't good enough for many calculations. In a variational quantum circuit, part of the application remains in the classical computer, which is responsible for calculating a certain value—a guess, a probable prediction, seeking the lowest energy value of the system, in this case, a Hamiltonian system. The value is generated and sent to the quantum computer, which runs its application and measures the quantum-processed value, sending it back to the classical computer. If the value is incorrect, the classical computer reduces its guess again, and the process continues, where the classical computer calculates and the quantum computer validates according to the model within the quantum circuit until it reaches the correct value. As expected in a continuous loop, the more shots and measurements taken, the greater the precision of the results tends to be. Currently, variational quantum circuits are the state of the art in quantum computing, possessing a hybrid problem-solving architecture. In the tests performed, they demanded the most total processing time for problem execution and showed low precision. For small datasets, they may not be the best processing choice due to their hybrid operating architecture and network latency. The tests were performed on a local computer in Ireland and a quantum computer in Washington, USA. Network latency and local processing on a notebook did not result in a performance gain in this processing. A good classical computer with Nvidia cards to accelerate calculations and a local network with the quantum computer with low latency can deliver much more significant results.

3.9 Classic reference models

For this study, four well-established classical algorithms were chosen as baseline models, using SciKit-Learn with fixed hyperparameters, as shown in the table below, for comparative tests with quantum algorithms, as shown in the table below.

Model	Type	Principal hyperparameters
SVM (RBF)	Maximum margin	kernel = RBF, C = 1.0, gamma = scale
k-NN	Based on neighborhood	k = 5, uniform weights, Euclidean distance
Logistic Regression	Linear	solver = lbfgs, C = 1.0, max_iter = MAX_ITER
MLP	Neural network	hidden layers (64, 32), ReLU activation, early stopping

Table 3.4 — Classic reference and configuration models.

3.10 Quantum models

The work includes seven quantum and hybrid models (quantum models using variational quantum circuits) for analysis, all executed in the same application execution context, and divided into three functional groups.

3.10.1 Variational Component

The models based on variational circuits combined three configurable parts as follows:

- **Feature maps:** ZZFeatureMap and PauliFeatureMap [16], [17], responsible for the non-linear encoding of data in Hilbert space;
- **Ansätze (parameterized circuits):** RealAmplitudes, EfficientSU2 and TwoLocal [4], with full entanglement;
- **Classical optimizers:** COBYLA [3], SPSA [18] and L-BFGS-B, ADAM in the variant register, responsible for adjusting the ansatz parameters to minimize the cost function.

3.10.2 Variational Models (QCL, VQC and QNN)

QCL (Quantum Circuit Learning) and VQC (Variational Quantum Classifier) [19], [16] use ZZFeatureMap together with RealAmplitudes, which are optimized by COBYLA, an optimizer used to train variational quantum models, being used to discover the best values of the adjustable circuit parameters. QNN (Quantum Neural Network) is built on a SamplerQNN with ansatz EfficientSU2 and a parity interpretation function that maps the measured bit strings into class labels. These models have an execution cost of $O(\text{MAX_ITER} \times n)$, as each iteration of the optimizer submits circuits for all samples.

3.10.3 Kernel and Fidelity Models (QSVM and QkNN)

QSVM employs a support vector machine whose kernel is the quantum fidelity call between encoded states, calculated by a fidelity circuit using ComputeUncompute [16], [17].

3.10.4 Non-variational Quantum Baselines (QFC and QFE-LR)

Below are quantum models compared with classical models, two of which can be called inexpensive models because they are used to compare with other variational quantum models that are trained by adjusting their circuits several times by trial and error. The QFC and QFE-LR models are used as test bases for variational circuits so that other variational circuits find the results that best fit the expected outcome.

Table 3.5 presents the seven quantum models.

Model	Classic analogue	Variational?	Characterization
QSVM	SVM	No (kernel)	Quantum fidelity kernel + SVC

Model	Classic analogue	Variational?	Characterization
QkNN	k-NN	No	Similarity by fidelity (or Gaussian proxy)
QCL	Logistics Reg.	Yes	VQC (ZZFeatureMap + RealAmplitudes, COBYLA)
VQC	MLP	Yes	Variational classifier, convergence capture
QNN	—	Yes	SamplerQNN (EfficientSU2 + parity)
QFC	—	No	Classifier by average fidelity (baseline)
QFE-LR	—	No	Fixed quantum embedding + logistic regression

Table 3.5 — Quantum models, classical analogues, and variational classification.

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Introduction

Here we discuss the results of running all the models, both classical and quantum machine learning, applied to the Iris dataset. All models ran in the same execution, generating data logs and finally generating execution graphs. The computer where the application ran is a Dell notebook with an 11th Generation Intel Core i7-1185G7 3.00GHz processor, 32 Gigabytes of RAM, and 1 Terabyte of SSD. The notebook, defined as the local machine and responsible for running part of the quantum application on real IBM quantum hardware, the `ibm_marrakesh` quantum computer, was automatically chosen by the application because it had the fewest tasks during execution. The execution generated sixteen graphs organized into four axes: hypothesis verification, recognition of limitations, and conclusions of the work. All numbers cited come from the structured results file (`comparison_results.json`) exported by the execution itself and available on the GitHub link in the document footer.

4.2 Experiment Summary

The experiment compared four classical models (SVM, k-NN, Logistic Regression, and MLP) with seven quantum and hybrid models (QSVM, QkNN, QCL, VQC, QNN, QFC, and QFE-LR), using the same dataset,

in this case Iris reduced to 2 principal components (2 qubits), with stratified partitioning and a small test subset of six samples, two per class, imposed by hardware guards. The quantum component was executed on the `ibm_marrakesh` superconducting QPU with 156 physical qubits, of which only 2 were used. The aggregated results had an average classical accuracy of 0.750 versus 0.357 for the quantum, and the total classical training time was 0.013 seconds, which was about five orders of magnitude less than the quantum time of 1.858 seconds.

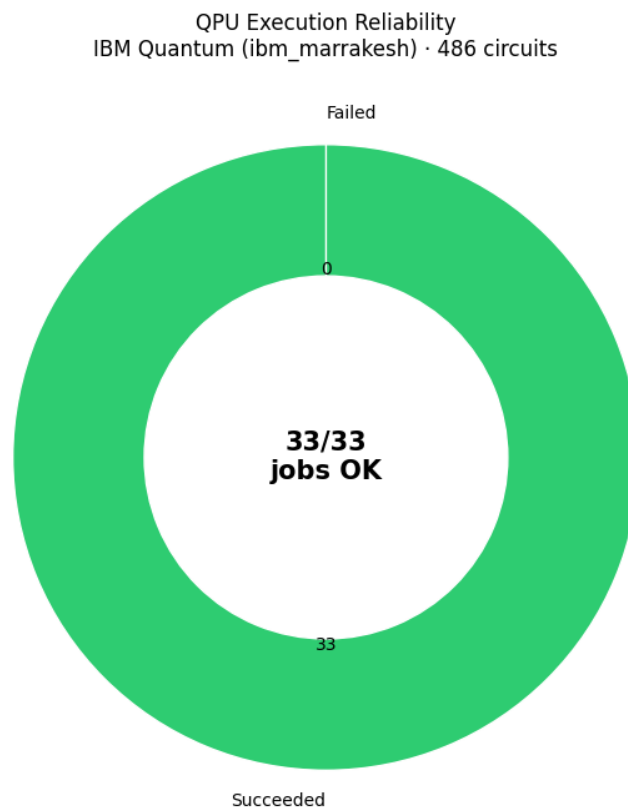


Figure 6 — QPU execution reliability: 33/33 jobs completed, 486 circuits, zero failures.

Below is a graph detailing the submission times of executions on the quantum computer, showing the number of quantum circuits per job. It's possible to identify a gap in the execution queue of 843 seconds between Jobs 15 and 16. It's important to note that the total time is 2,227 seconds, but a large part of this time is related to waiting time in the queue, as quantum computers are shared by several users in the cloud. The use of QPUs is presented throughout the chapter, and although the time shown in this execution gap is impressive, it does not represent the internal execution time of the QPU and requires caution in the analysis.

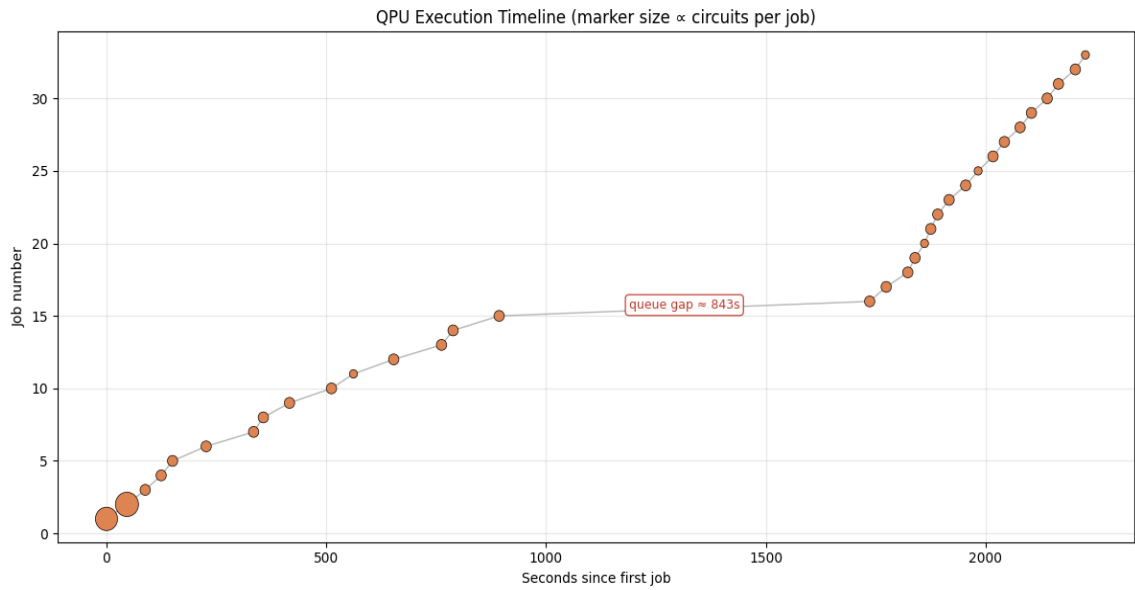


Figure 7 — Execution timeline in the QPU: the queue gap of ~843 s dominates the wall time.

The figure below shows the same analysis for the circuits submitted and presents the same pattern of behavior. The total number of circuits used in the complete execution reached 486 quantum circuits with a significant execution queue latency characteristic of shared quantum resources.

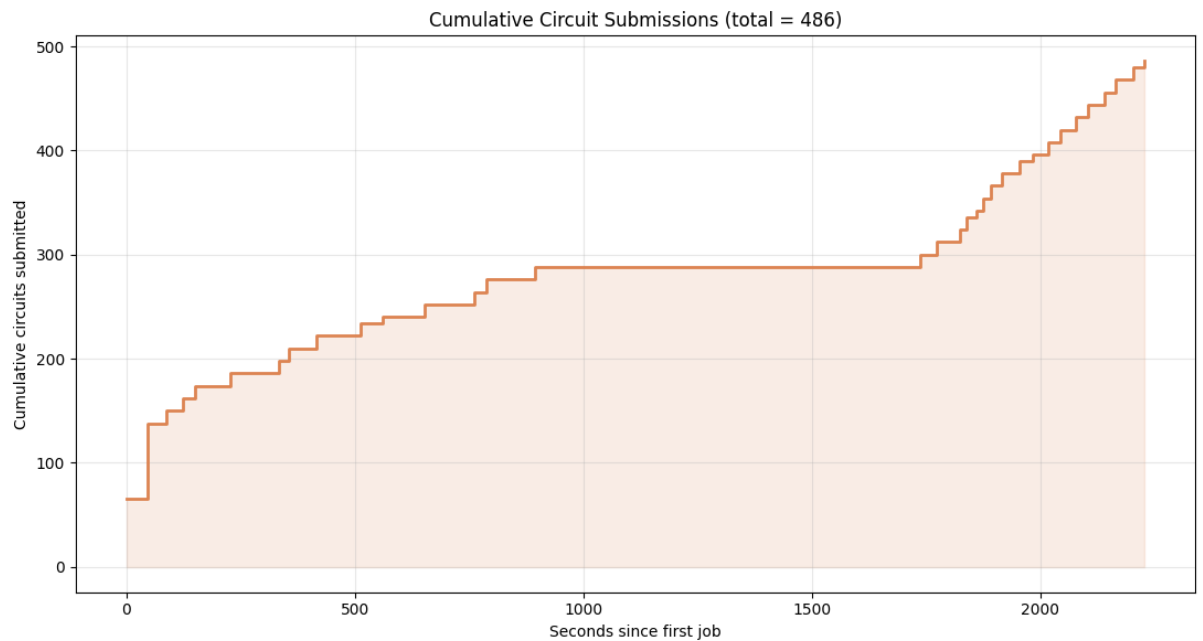


Figure 8 — Circuits accumulated over time: the central plateau reflects queuing.

4.3 Predictive Performance: Accuracy and Metrics

Figure 9 compares accuracy by classical × quantum pair; in all pairs, the classical model equals or surpasses the quantum model. The highlight is Logistic Regression, which achieves perfect accuracy

(1.000) on the test set, while its quantum pair (QCL) is 0.333. The only tie occurs in k-NN<->QkNN (both 0.667) - a result that requires caveat, discussed later.

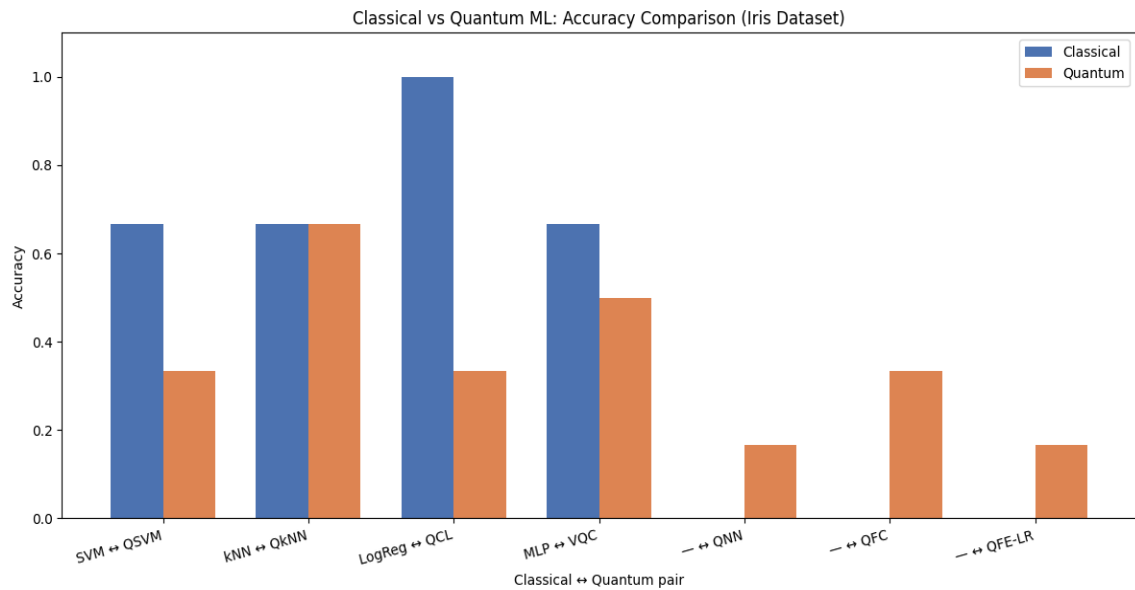


Figure 9 — Accuracy by classical × quantum pair on Iris: the classical pair equals or surpasses in all cases.

Table 4.1 consolidates the metrics of all models, extracted directly from the results.

Model	Paradigm	Accuracy	F1-macro	Training(s)	Depth	In QPU?
LogReg	Classic	1,000	1,000	0,003	—	—
SVM	Classic	0,667	0,556	0,001	—	—
kNN	Classic	0,667	0,556	0,001	—	—
MLP	Classic	0,667	0,556	0,008	—	—
QkNN	Quantum (n-var)	0,667	0,556	~0,000	0	No (proxy)
VQC	Quantum (var)	0,500	0,389	1207,3	2	Yes
QSVM	Quantum (n-var)	0,333	0,333	46,7	0	Yes
QCL	Quantum (var)	0,333	0,278	473,7	2	Yes

Model	Paradigm	Accuracy	F1-macro	Training(s)	Depth	In QPU?
QFC	Quantum (n-var)	0,333	0,333	~0,000	0	Yes
QNN	Quantum (var)	0,167	0,133	107,8	2	Yes
QFE-LR	Quantum (n-var)	0,167	0,111	22,9	0	Yes

Table 4.1 — Metrics and cost per model (n-var = non-variational; var = variational).

It is important to make a caveat, QkNN presented the best performance of the quantum models by accuracy (0.667) registering `ran_on_backend = false`, that is, it fell into the classical Gaussian kernel proxy predicted in the configuration, not having been executed on the QPU. As for the models that ran 100% on quantum hardware, the one that presented the best performance was VQC (0.500).

The figure below shows the errors of the best classical and quantum models in Figure 10, where Logistic Regression produces a perfect diagonal (2/2/2). QkNN correctly identifies Setosa and Versicolor, but classifies all samples from Virginica as Versicolor. This error was mentioned in section 4.2.2 due to the use of PCA which reduced the number of dimensions from 4 to 2 dimensions. The objective of using PCA was to reduce the number of Qubits to 2 in order to reduce the noise level accompanied by an increase in complexity. The use of PCA brought the Versicolor and Virginica classes closer together, however it penalized other models, so much so that later in the confusion matrix we can see that QkNN confused Virginica with Versicolor.

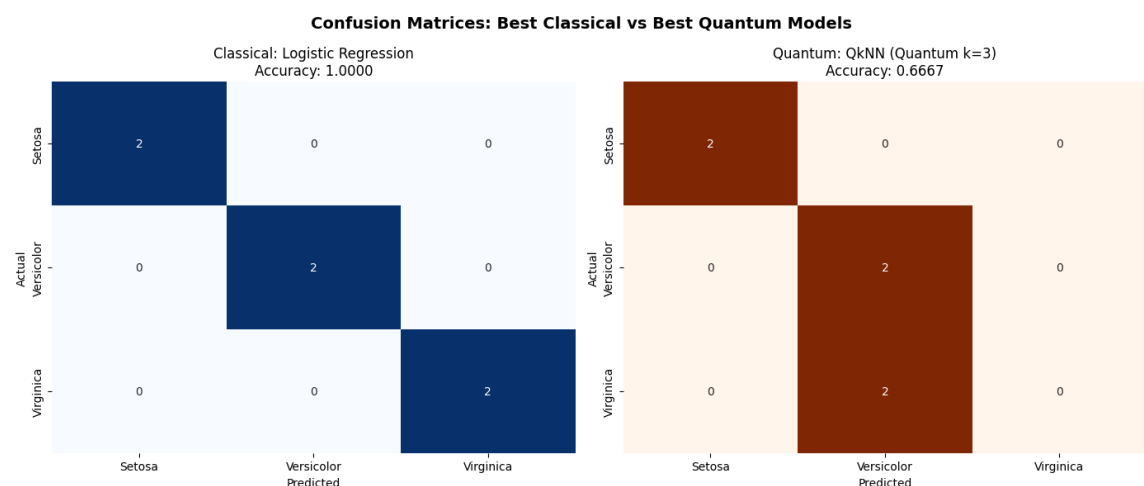


Figure 10 — Confusion matrices: LogReg (perfect diagonal) vs QkNN (confuses Virginica with Versicolor).

The figure below shows a radar and just below that a heat map analyzing accuracy, F1-macro, precision and recall. Logistic Regression fills the maximum polygon in all directions, while the quantum best occupies a noticeably smaller area. The heat map also presents the worst performances, which are QNN and QFE-LR, with scores between 0.083 and 0.167, and confirms the homogeneity of the SVM/kNN/MLP

trio at 0.667.

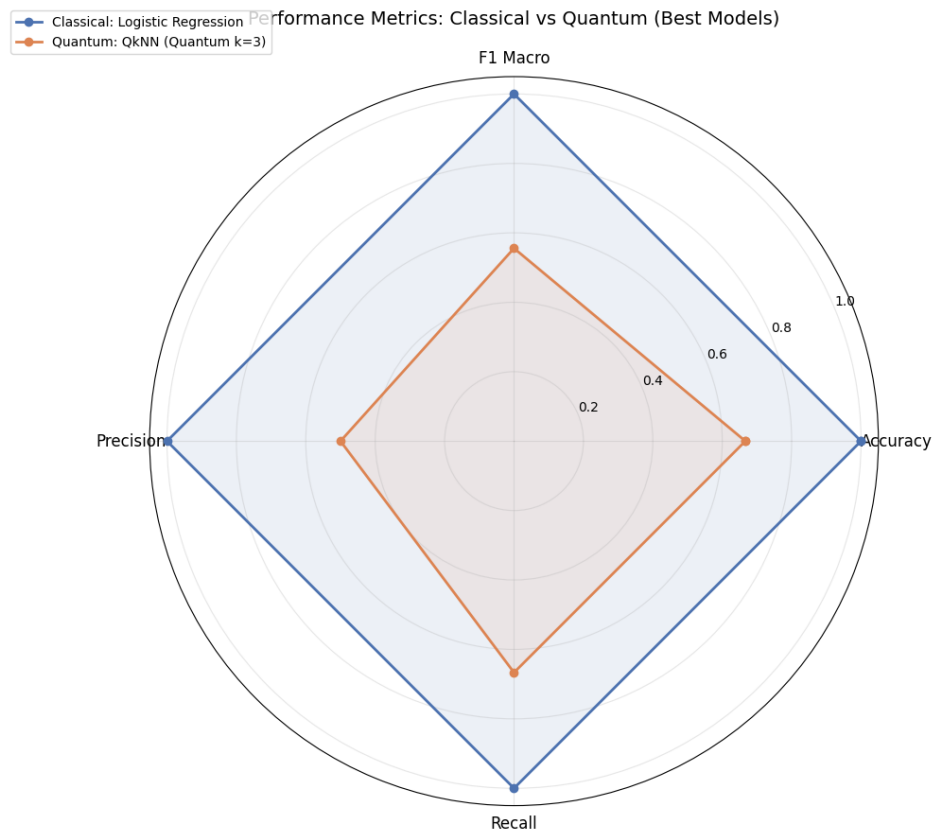


Figure 11 — Radar of metrics for the best models: LogReg dominates QkNN in all dimensions.

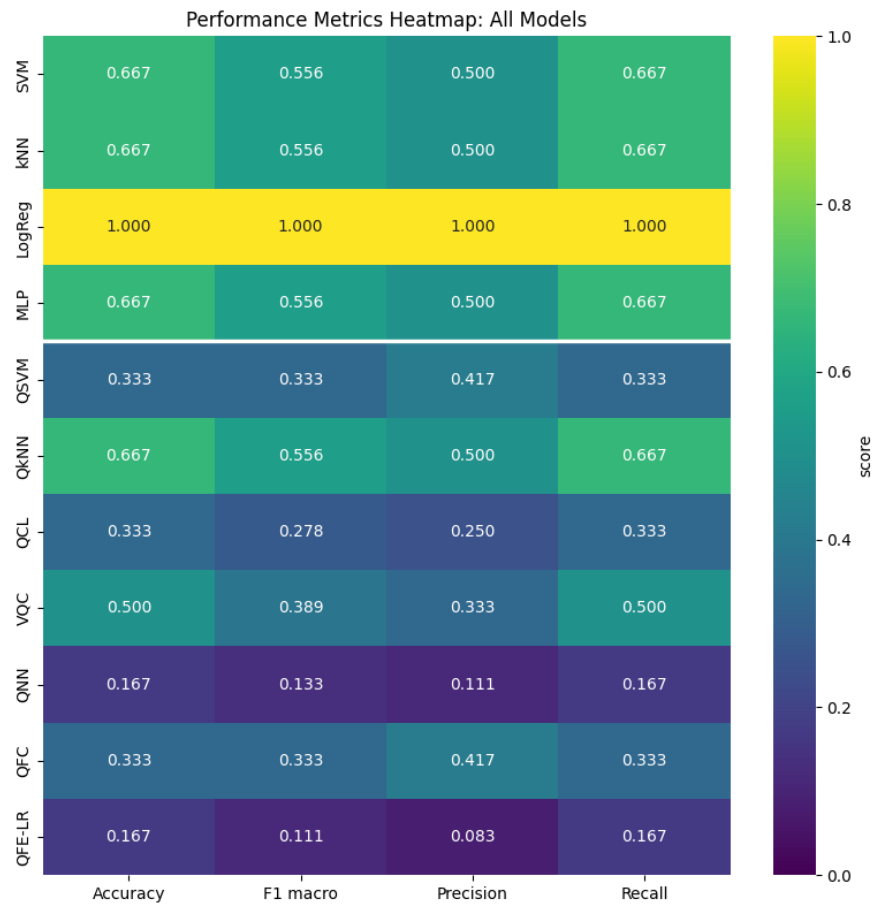


Figure 12 — Heat map of all metrics and models: LogReg at the top, QNN/QFE-LR at the bottom.

4.4 Computational cost: the price of quantum computing in the NISQ era

According to the graphs presented, the accuracy favors the processing of classical models running on a local machine, making the computational cost very high. The figure below shows the accuracy vs. training time plane, where the models are concentrated in the upper left corner, in the case of high accuracy and negligible time, and in contrast, the quantum models running on the quantum computer are spread to the right and down, with the VQC isolated in ~1.207 s of training.

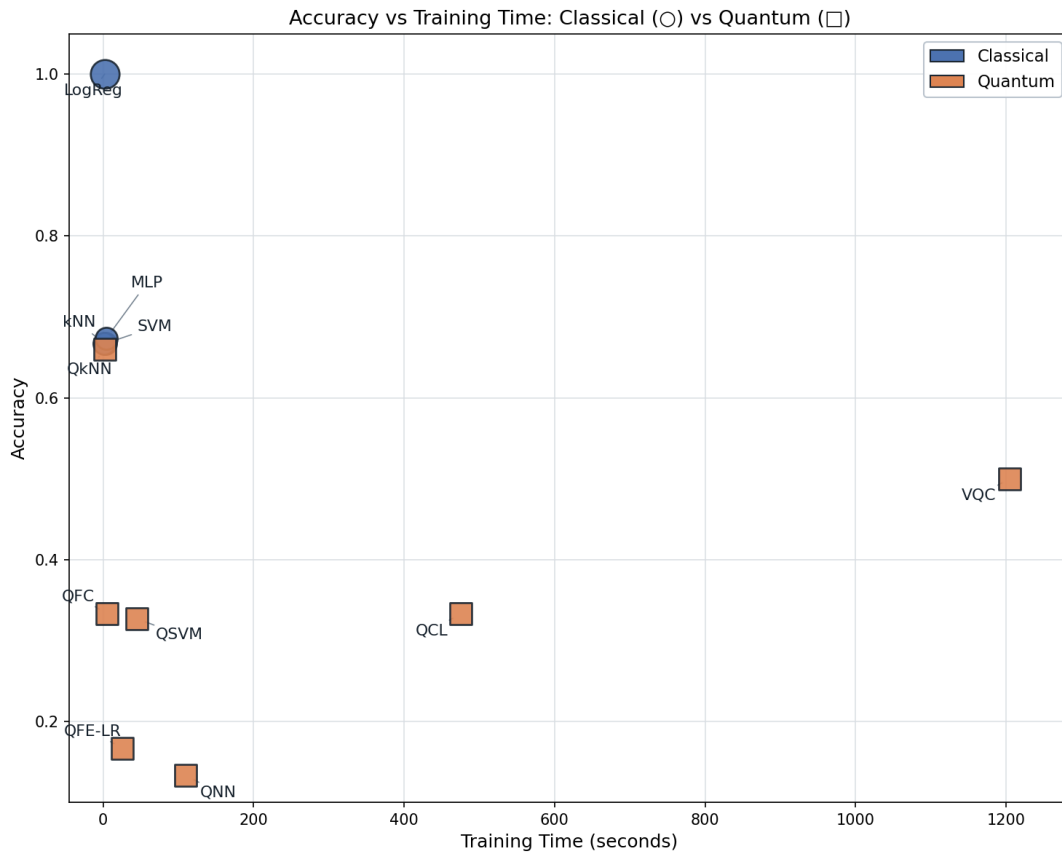


Figure 13 — Accuracy × training time: the classical models occupy the ideal corner (high accuracy, low cost).

The two figures below, on a logarithmic scale, show the training and inference points. Classical models run in milliseconds, while variational models run in thousands of seconds. In inference, differences persist in QFC (Quality Classification Classification) at 187 seconds, being a fidelity classifier that recalculates similarities in hardware with each prediction, while QCL (Quality Classification Classification) achieves 91 seconds, compared to fractions of a second for classical models.

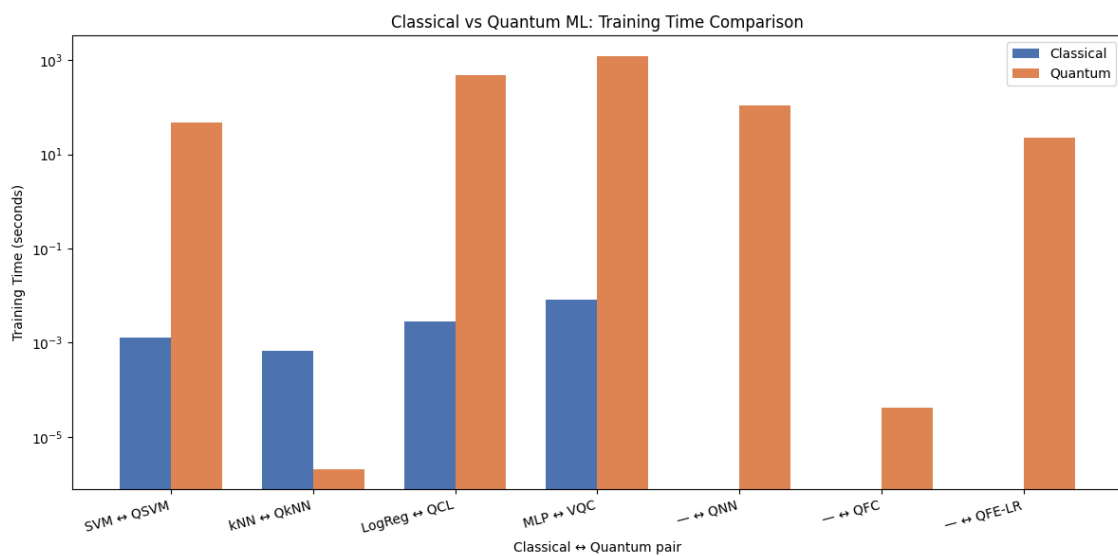


Figure 14 — Training time per pair (log scale): quantum computing costs orders of magnitude more.

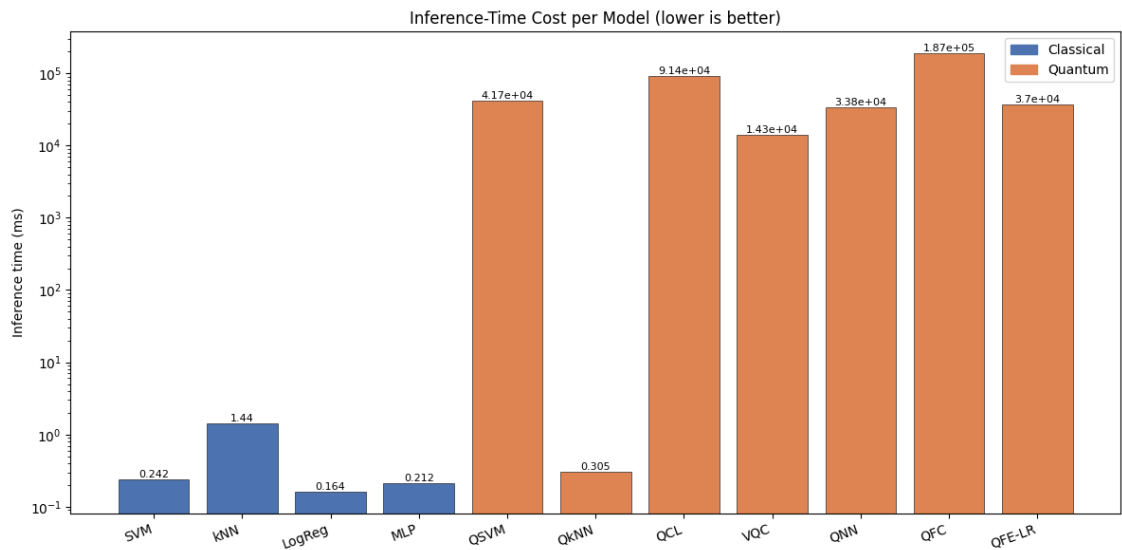


Figure 15 — Cost of inference per model (log scale): QFC and QCL are the most expensive.

The figure below quantifies the over cost as a quantum/classical time ratio per pair. The factors are extreme, being 36.721x for SVM→QSVM, 146.942x for MLP→VQC, and 168.759x for LogReg→QCL. The only pair where quantum appears to be faster, in the case of k-NN→QkNN, is again the case of the classical proxy, and does not represent real quantum execution, which can be seen in the next table detailing these pairs.

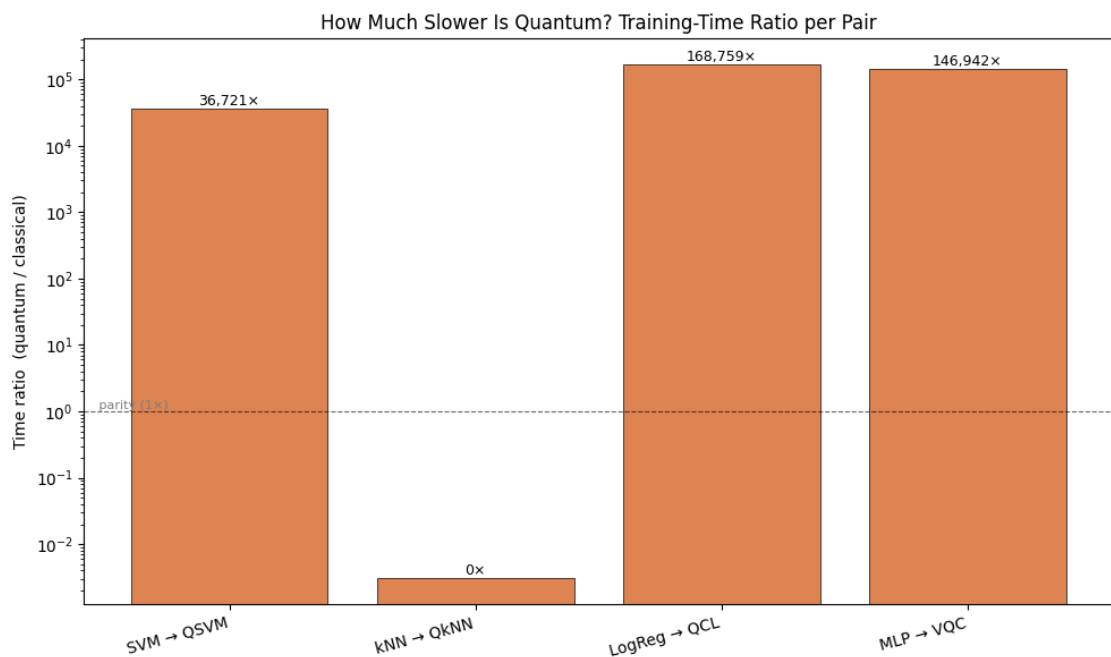


Figure 16 — Training time ratio (quantum/classical) per pair: factors of 10^4 – 10^5 x.

Pair (classical -> quantum)	Classic Accuracy	Quantum accuracy	Gap	Time ratio
SVM → QSVM	0,667	0,333	0,333	36.721×
kNN → QkNN	0,667	0,667	0,000	0,003× (proxy)
LogReg → QCL	1,000	0,333	0,667	168.759×
MLP → VQC	0,667	0,500	0,167	146.942×

Table 4.2 — Paired comparison of accuracy and training time.

4.5 Variational models: convergence, depth, and overhead

The figure below shows the reason for the low variational performance; the convergence curves are flat [20]. For QCL and VQC, the objective function varies between 11.3 and 12.0 in eight iterations without a consistent drop. For QNN, it remains at 1.25, which is consistent with the iteration limit imposed by the hardware guards. Hardware guards can be defined as the safety limits of the project that are automatically activated when it detects an execution that will run on a QPU instead of the simulator. They allow for cost and time control. Most hardware guards are in the ".env" configuration file, such as the COBYLA iteration number, which limits it to 8 rounds. Looking at the variational models, QCL, VQC, and QNN, we can see that they had poor performance, which does not refer to any kind of failure in quantum processing, but because the hardware guard performed a short training to save QPU time.

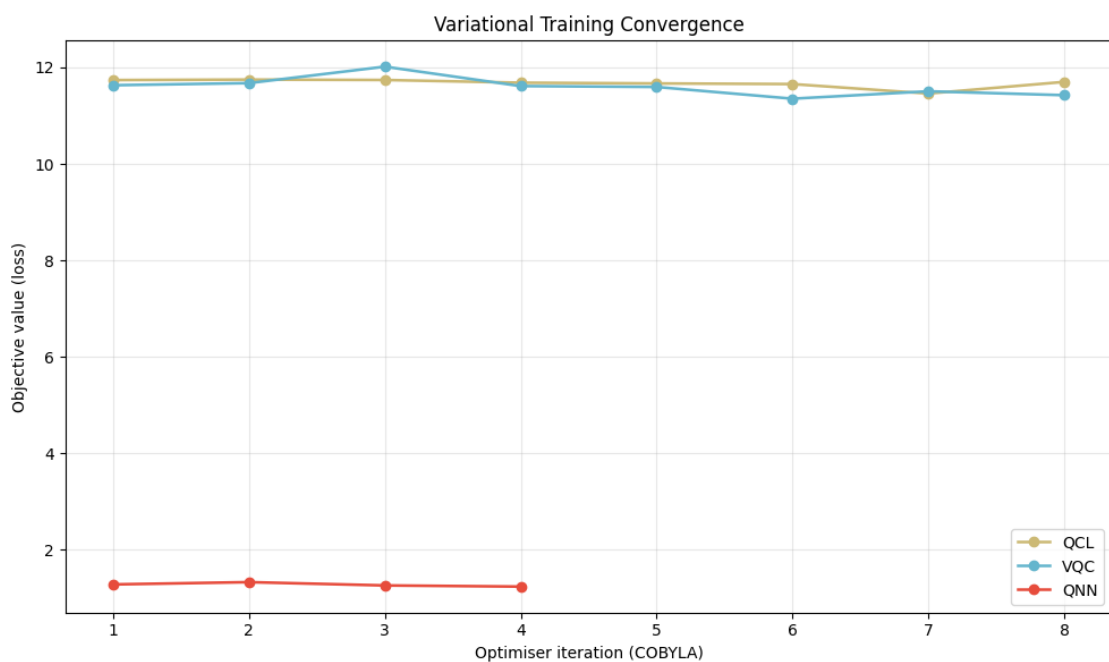


Figure 17 — Variational convergence: planar curves under the 8-iteration ceiling (COBYLA).

The two figures below illustrate the depth of quantum circuits. Every time a code is executed, a transpilation process is triggered that transforms the model into a quantum circuit, in this case a Qiskit transpilation to QASM (Quantum Assembler). In simple terms, depth represents how long a quantum circuit is due to the complexity of its operation; the longer the circuit, the more quantum gates are used. The variational models QCL, VQC, and QNN exhibit the greatest depth, which translates to longer execution times and greater complexity. The second figure below demonstrates that depth did not translate to greater accuracy in this experiment. The QkNN with depth 0 showed an accuracy of 0.667, being the most precise, while QNN with depth 2 showed an accuracy of 0.167, the worst. In this case, more circuits in the experiment did not translate to greater accuracy.

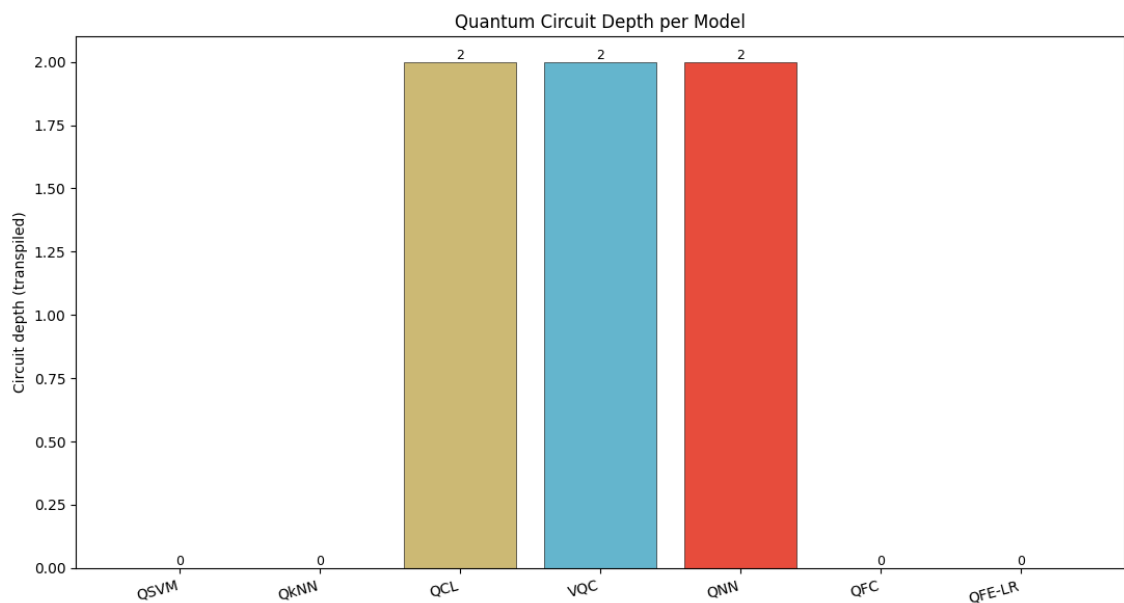


Figure 18 — Circuit depth by model: only variational models have a depth of 2.

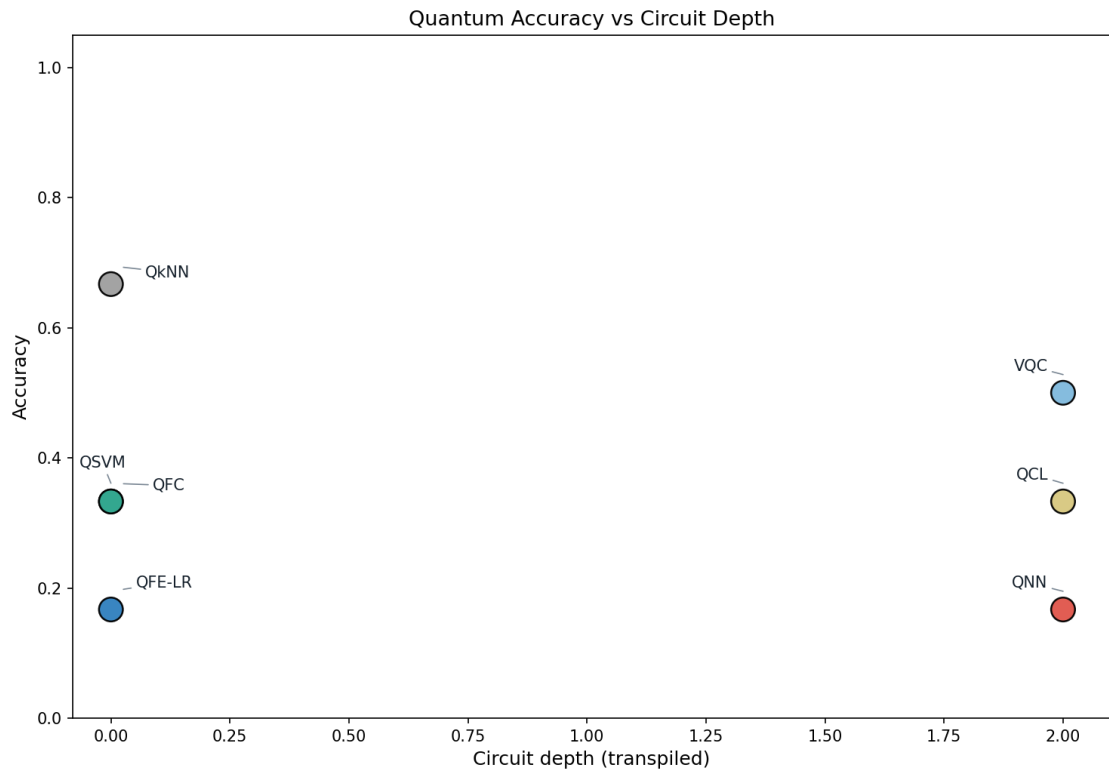


Figure 19 — Accuracy $\times$ depth: no correlation, depth does not explain performance.

Below, the figure shows a comparison separating quantum models into variational and non-variational models regarding total computation time, training + inference. The non-variational average is 83.9 seconds, while the variational average is 643 seconds, proving that variational quantum models are 7.7 times slower. The QFC model, which is non-variational, has a very high inference time of 187 seconds, breaking the notion that quantum cost is not reducible to the variational/non-variational dichotomy.

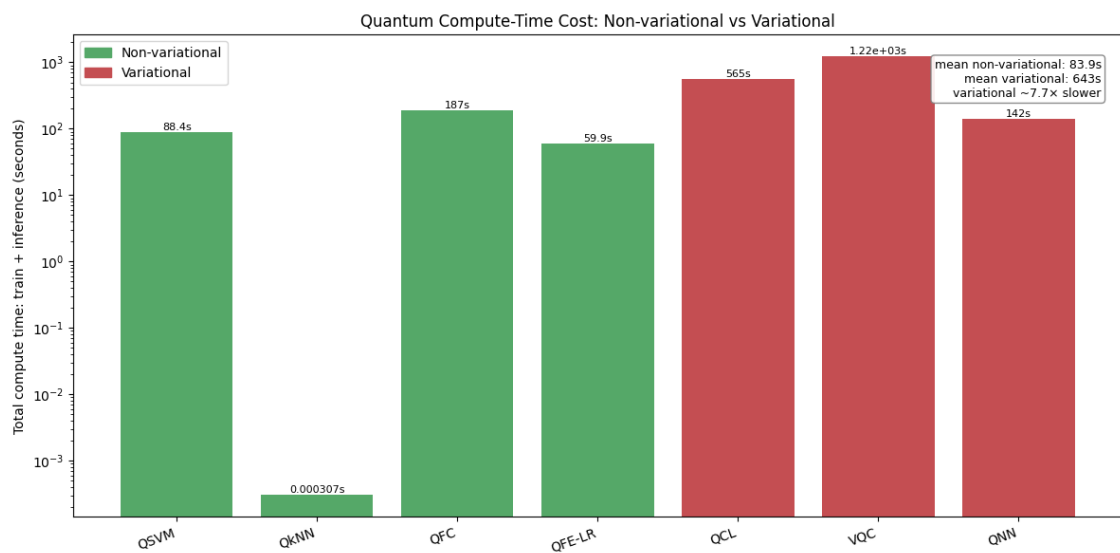


Figure 20 — Computational cost: variational models $\sim 7.7\times$ slower than non-variational (average).

4.6 Classical vs. Quantum Paired Analysis

The figure below shows the absolute difference per pair. The largest gap is in LogRef->QCL with a factor of +0.667 in favor of the classical, followed by SVM->QSVM with a factor of +0.333 and MLP->VQC with a factor of +0.167. The k-NN->QkNN pair shows a zero gap, but, as observed, the quantum side corresponds to the classical proxy. The classical proxy works by comparison/similarity or by a fixed circuit, without this repeated adjustment process. In summary, the classical wins three pairs and ties one, and the tie is not a genuine quantum execution.

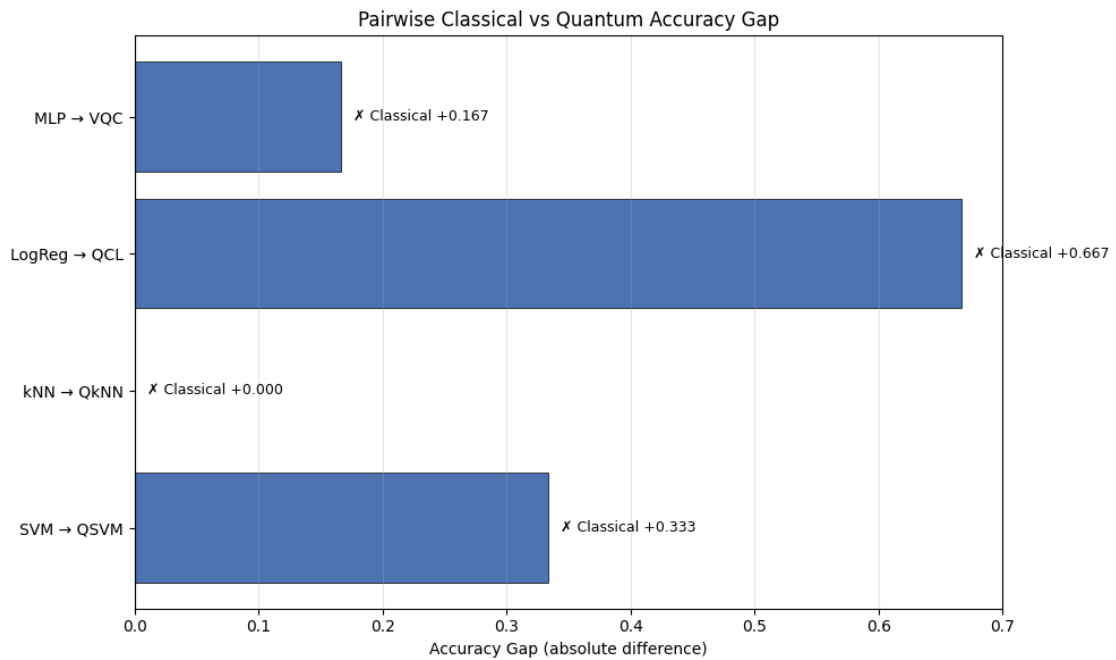


Figure 21 — Difference in accuracy per pair: the classical model wins or ties in all cases.

CHAPTER 5: CONCLUSIONS AND FUTURE WORK

Analysing the hypotheses formulated in the methodology, the null hypothesis (H_0), given the absence of an appreciable difference between the classical and quantum paradigms, is rejected. The average accuracy difference of 0.750 versus 0.357 and the latent time difference of up to $\sim 169,000x$ is significant. The alternative hypothesis (H_1) is corroborated; there is a performance difference attributable to the paradigm and the execution conditions. Considering the direction and cause, the observed quantum disadvantage, however, stems from the combination of NISQ noise, the eight-iteration ceiling for variational models, and the reduction of the problem to two dimensions, and not from a related impossibility of quantum methods.

The results should be read as a controlled case study, not as proof of the general superiority of any paradigm. The main limitations are:

- **Small test set:** only six samples (two per class), which makes each hit/miss account for ~ 0.167 accuracy and drastically reduces statistical power.
- **Dimensional reduction:** the PCA for 2 components discards information and approximates the Versicolor and Virginica classes, penalizing all models.
- **Iteration ceiling:** hardware guards limited COBYLA to eight iterations, preventing variational convergence.
- **Wall time:** quantum times include queuing and network latency (gap of ~ 843 s), not representing pure algorithmic time in the QPU.
- **QkNN proxy:** the best quantum result was not run in hardware and should be interpreted as a classical reference.

The aim of this work was to compare, in a reproducible (source code available on github) and verifiable way, classical and quantum models by running classical models on a low-platform system and quantum models for machine learning using the Iris dataset. Each execution in QML (Quantum Machine Learning) presented evidence such as logs and recorded data, as well as graphs generated at runtime, with auditable identifiers on the `ibm_marrakesh` computer. The conclusion is unequivocal regarding accuracy results between classical and quantum models. Where classical models showed superior accuracy, being more efficient, quantum models had an extreme computational cost with factors of 10^4 to 10^5x , without a proportional return in predictive quality.

It is possible to say, based on the methodology used, that the results confirm that the quantum advantage did not show significant results in small problems, such as the example used, the Iris dataset. The cost issue makes this type of use completely impractical, with a cost of US\$100.00 per minute, we can say that the processing time per job is very expensive. It would be very interesting to study other possibilities in more depth if there were greater availability of hardware so that other types of experiments could be performed.

Variational quantum circuits are currently the state of the art in error correction and precision improvement of NISQs; however, one of the problems encountered was the discrepancy in processing time. Although there is more quantum circuits involved in this processing, what made it most difficult to understand the reason for this unique performance was the lack of clear documentation about them. Most of the documentation related to variational quantum circuits is related to quantum physical and quantum information characteristics, but it is not clear in the literature what their architecture is, as they are hybrid models running on both quantum computers and classical computers during task execution runtime. Network latency and execution on a classical computer with limited numerical processing power created a significant performance discrepancy. For future researchers, using this type of quantum architecture requires a pre-evaluation of the performance of the classical computer being used for integration with the quantum computer. Qiskit's own packages handle this hybrid integration between classical and quantum platforms. However, it is recommended that the application running on the classical machine use high-performance Nvidia graphics cards or servers capable of handling a high-performance numerical processing architecture.

Another important aspect that was highlighted in the programming of the source code for the quantum machine was the number of qubits, generally 2 qubits, due to the very nature of machine learning models that worked with independent and dependent variables, with 1 qubit for each. This type of quantum circuit design greatly limits the performance of quantum computers, as it makes very limited use of the main computational resource of these quantum machines, which is the number of qubits. The greater the number of qubits used, the greater the possibility of working with multiple variables in a state of quantum entanglement, which is where the computational leap of quantum computers lies.

Considering the small number of qubits used in the tests and the execution characteristic of quantum computers, which is the number of shots (nothing more than the number of rounds performed, around 1024 for each quantum machine learning model), this represents a performance gap. While a classical computer runs the models in a single round, a quantum computer must perform several rounds to obtain an accurate result. In the tests with 1024 shots, most results showed low precision compared to the classical models. This could be improved by increasing the number of shots, as quantum machines perform measurements after each shot (round), generating a wave collapse effect where many results in a quantum state result in a single result. Increasing the number of shots provides greater precision and, on average, reduces the effects generated by noise from quantum machines. This makes it very difficult to believe that a quantitative machine learning classification or regression model can have significant commercial results, considering the limited number of qubits used, the number of shots required for satisfactory precision, the processing time compared to a simple classical machine, and finally the execution cost. IBM charges US\$100.00 per minute of execution. In our tests, we used the Iris dataset, which is very small. We cannot test larger datasets to analyse performance on large datasets, but evidently, small datasets can run very well on classical machines at fractions of the cost of 1 minute of a quantum computer.

HSBC's announcement [21] about significant financial results in the stock market using quantum computers for investment precision did not present details of the implementation used in IBM's quantum computers, which does not allow verification of these performance and precision results, but it is safe to say, based on what the work presented, that it was not simple models like those presented in this work using only 2 Qubits that gave the bank this advantage margin. The models used must have used much more complex models that involved dozens or hundreds of Qubits running in each model to exploit the maximum computational performance of quantum entanglement.

The use of PCA for component reduction was not a choice by a standard definition, so much so that there is no right or wrong for its use or not, but the increase in complexity could lead to excessive processing time in a 10-minute window of use of IBM's quantum computers. One of the problems presented in the QkNN confusion table showed model confusion between Virginia and Versicolor. It would be interesting to run the experiment again without PCA so that we could run with more qubits and re-analyze the results to compare them. We could also see if, in the case of a small dataset like the Iris dataset, reducing the number of components or qubits from 4 to 2 would represent a significant difference.

APPENDICES

Source Code of **IRIS_QML** in: https://github.com/wevertonsc/TUS DISSERTATION_2026

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